DAY 2 · TUESDAY 17 NOVEMBER 2026

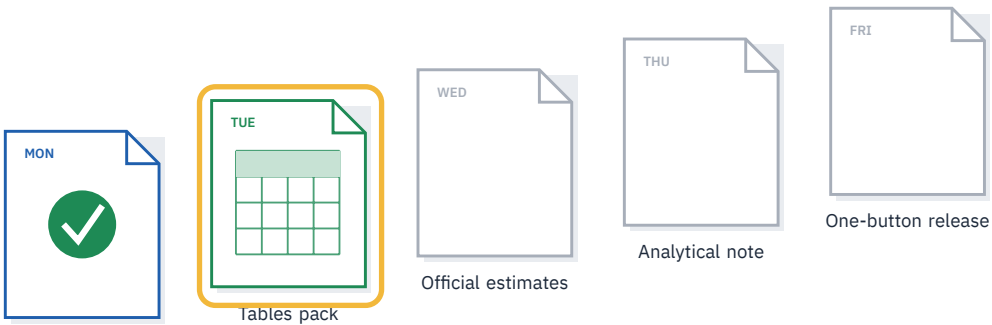
From monthly files to the tables pack

Data integration, descriptive analysis and reporting



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Five days, five products



Today’s files are already on your PC.

YOUR TURN · TASK 1

Today’s files

DO

Type:

```
dir "C:/NBS_Stata/day2"
```

YOU SHOULD SEE

The day’s files: two labs, the practice set, the daily check and the `catchup` folder.

IF NOT

Nothing listed? Copy the `day2` folder from the USB stick into `C:\NBS_Stata`.

Yesterday’s hardest point: *filled from the Day 1 check*.

The fix, on yesterday’s code

```
the corrected line, filled from the Day 1 check
```

YOUR TURN · TASK 2

Rebuild yesterday's file

DO

1. Open `day1/my_day1.do`.
2. Press **Ctrl + D**.

YOU SHOULD SEE

The whole of Day 1 runs again and ends with `file data/lfs_teach.dta saved`.

IF NOT

Red somewhere? Open `catchup_D1_S9.do` instead and run it.



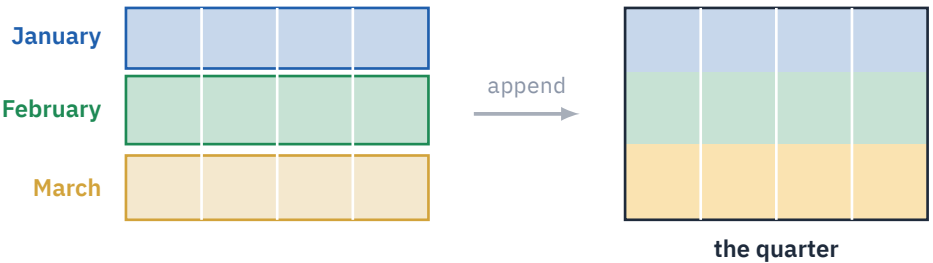
1

DAY 2 · PART 1

Three months, one file

NBS interviews every month. The quarter is built by stacking the months.

Same columns, more rows



YOUR TURN · TASK 3

Open January

DO

Type:

```
use "data/lfs_jan.dta", clear
count
```

YOU SHOULD SEE

One number: the persons interviewed in January.

IF NOT

`file not found`? Check that `cd` points to `C:/NBS_Stata`.

EXTRA `describe, short`: variables, observations and size in three lines.

Your result

```
. cd "C:/NBS_Stata"
C:\NBS_Stata

. use "data/lfs_jan.dta", clear

. count
  1,864

. describe, short

Contains data from data/lfs_jan.dta
 Observations:         1,864
    Variables:            90                  1 Oct 2026 21:10
Sorted by: unique_ind

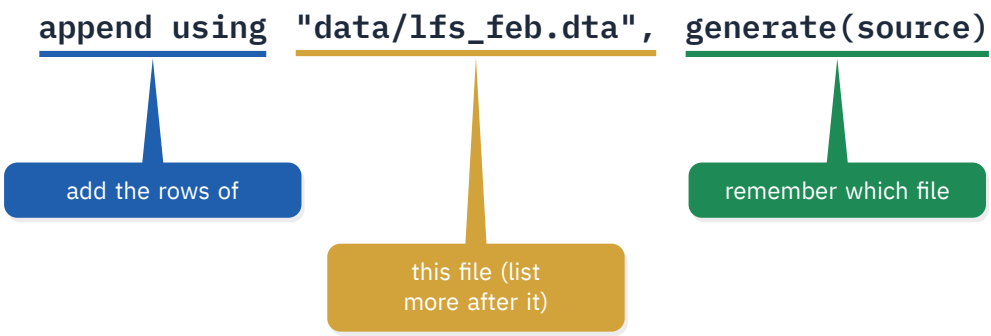
.
```

Three months, three counts

```
. use "data/lfs_jan.dta", clear
. count
  1,864 (1)
. use "data/lfs_feb.dta", clear
. count
  2,082 (2)
. use "data/lfs_mar.dta", clear
. count
  1,942 (3)
```

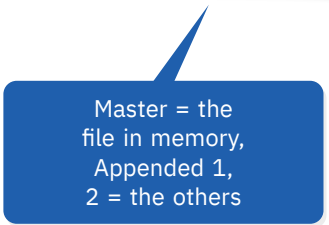
1. January: 1,864 persons
2. February: 2,082
3. March: 1,942
4. together they must make the quarter: 5,888

Stack files under each other:



Remember where each row came from

```
append using "data/lfs_feb.dta" "data/lfs_mar.dta", generate(source)
```



Here `int_month` tells the same story; in most NBS files nothing does.

YOUR TURN · TASK 4

Build the quarter

DO

Add to `my_day2.do` and run:

```
use "data/lfs_jan.dta", clear
append using "data/lfs_feb.dta" ///
  "data/lfs_mar.dta", generate(source)
count

tabulate source
```

YOU SHOULD SEE

Many lines (`label … already defined`): normal. Then the quarter's count and `source`: Master, Appended dataset 1, 2.

IF NOT

`file not found`? Check each file name: `lfs_jan`, `lfs_feb`, `lfs_mar`.

Your result

```
(label ilo_dis_lab already defined)
(label neet_lab already defined)
(label sub_all_lab already defined)

.
.    count
  5,888

.
.    tabulate source
```

Dataset source	Freq.	Percent	Cum.
Master	1,864	31.66	31.66
Appended dataset 1	2,082	35.36	67.02
Appended dataset 2	1,942	32.98	100.00
Total	5,888	100.00	

The notes append prints, and why they are harmless

```
. use "data/lfs_jan.dta", clear

. append using "data/lfs_feb.dta" "data/lfs_mar.dta", generate(source)

(variable ilo_job1_how_usual was byte, now int to accommodate using data's values)
(variable ilo_joball_how_usual was byte, now int to accommodate using data's values)

(label sub_all_lab already defined)
(label neet_lab already defined)
(label ilo_dis_lab already defined)
(label unemp_aggr already defined)

. count
  5,888
```

1. February has a larger value: Stata widens the storage type to fit it
2. value labels already in January are kept; about 70 such lines follow
3. the count you check: 5,888

Where each row came from

```
. tabulate source

    Dataset source |      Freq.     Percent        Cum.
-------------------+-----------------------------------
            Master |      1,864       31.66       31.66
Appended dataset 1 |      2,082       35.36       67.02
Appended dataset 2 |      1,942       32.98      100.00
-------------------+-----------------------------------
             Total |      5,888      100.00
```

1. Master: the file in memory, January
2. Appended dataset 1 and 2: February and March
3. 5,888: the quarter

Does each file hold its own month?

```
. tabulate source int_month

                   |     Interview month, Q1 2026
    Dataset source |   January   February      March |     Total
-------------------+---------------------------------+----------
            Master |     1,864          0          0 |     1,864
Appended dataset 1 |         0      2,082          0 |     2,082
Appended dataset 2 |         0          0      1,942 |     1,942
-------------------+---------------------------------+----------
             Total |     1,864      2,082      1,942 |     5,888
```

1. January file: only January interviews
2. February file: only February
3. zeros off the diagonal: no interview filed under the wrong month

QUICK CHECK

After append, Stata prints “variable ilo_job1_how_usual was byte, now int. What do you do?

A. nothing: Stata made room for a larger value

B. stop: the data are damaged

C. run append again

ANSWER

After append, Stata prints “variable ilo_job1_how_usual was byte, now int. What do you do?

A nothing: Stata made room for a larger value

B stop: the data are damaged

C run append again

A `byte` holds whole numbers up to 100. February has a usual-hours value above that, so Stata switched to `int`. No value changed.

YOUR TURN · TASK 5

Check the quarter

DO

Add and run:

```
assert _N == 5888
```

YOU SHOULD SEE

Nothing: the quarter has the same 5,888 persons as yesterday's clean file.

IF NOT

`assertion is false`? A month was appended twice: rerun the file from the top.

YOUR TURN · TASK 6

Which districts in which month?

DO

Type:

```
tabulate district_q int_month
```

YOU SHOULD SEE

A table: 26 districts down, 3 months across.

IF NOT

Too wide? Make the Results window wider, or use the Extra.

EXTRA `tabulate int_month`: persons per month.

COMMON ERROR

A check that fails

```
. assert _N == 5800
assertion is false
```

WHY

The quarter has 5,888 rows, not 5,800. `assert` stops the do-file at this line, before anything is built on a wrong file.

FIX

Correct the number in the assert, or find the missing month: never delete the check.

COMMON ERROR

A file that is not there

```
. append using "data/lfs_apr.dta"
file data/lfs_apr.dta not found
```

WHY

There is no April file: the name or the folder is wrong.

FIX

Check the name in File Explorer; use the short path from `C:/NBS_Stata`.

The quarter in four lines

```
. describe, short

Contains data from data/lfs_jan.dta
 Observations:         5,888 (1)
    Variables:            91                  1 Oct 2026 21:10 (2)
Sorted by:
     Note: Dataset has changed since last saved. (3)
```

(1) 5,888 observations

(2) 91 variables: the 90 of a month file plus `source`

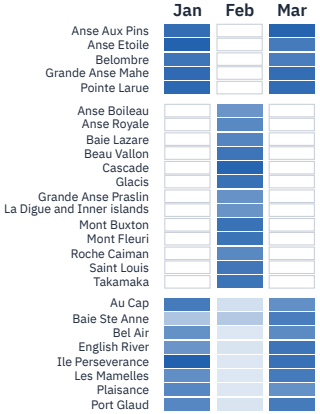
(3) the file in memory is new: save it before you lose it (Task 8)

Your result

```
. tabulate district_q int_month
```

DISTRICT	Interview month, Q1 2026 January	February	March	Total
English River	118	31	155	304
Mont Buxton	0	158	0	158
Saint Louis	0	153	0	153
Bel Air	124	10	132	266
Mont Fleuri	0	157	0	157
Plaisance	135	5	124	264
Roche Caiman	0	133	0	133
Les Mamelles	128	10	150	288
Ile Perseverance	178	23	161	362
Cascade	0	172	0	172
Pointe Larue	170	0	166	336
Anse Aux Pins	164	0	175	339
Anse Royale	0	133	0	133
Takamaka	0	157	0	157
Au Cap	150	36	126	312
Baie Lazare	0	136	0	136
Anse Boileau	0	119	0	119
Grande Anse Mahe	167	0	163	330
Port Glaud	135	30	150	315
Belombre	155	0	145	300
Beau Vallon	0	155	0	155
Glacis	0	160	0	160
Anse Etoile	176	0	149	325
Grande Anse Praslin	0	123	0	123
Baie Ste Anne	64	61	146	271
La Digue and Inner is	0	120	0	120
Total	1,864	2,082	1,942	5,888

The rotation, read from the data



5 districts: January and March

13 districts: February only

8 districts: January and March, with some interviews in February

This is the LFS design you run in the field, read back from the data.

RULE

append matches columns by name.

A variable renamed in one month becomes a new column, empty in the other months.

Check names before appending: `describe` each file.

YOUR TURN · TASK 7

See what a renamed variable does

DO

In the Command window, not in the do-file:

```
use "data/lfs_mar.dta", clear
rename ilo_sex sex
save "data/test_mar.dta", replace
use "data/lfs_jan.dta", clear
append using "data/test_mar.dta"
tabulate ilo_sex, missing
```

YOU SHOULD SEE

A missing row in `ilo_sex`: the March records, whose sex is now in `sex`.

IF NOT

No missing row? Check that the `rename` line ran.

One renamed variable, half a column empty

```
. tabulate ilo_sex, missing

        Sex |      Freq.     Percent        Cum.
------------+-----------------------------------
   1 - Male |        880       23.12       23.12
 2 - Female |        984       25.85       48.98
          . |      1,942       51.02      100.00
------------+-----------------------------------
      Total |      3,806      100.00
```

1

1. 1,942 missing: every March record, whose sex is now in sex
2. append matched by name: `ilo_sex` and `sex` are two different variables

ON YOUR OWN · A

January and March only

THE QUESTION

Build a file with the January and March interviews only. How many persons? Check the months with a table.

HINT

One `use`, one `append` with one file name, then `count` and `tabulate int_month`.

ON YOUR OWN · A · ANSWER

```
use "data/lfs_jan.dta", clear
append using "data/lfs_mar.dta", generate(source)
count
tabulate int_month
```

```
. count
  3,806 (1)

. tabulate int_month

  Interview |
  month, Q1 |
       2026 |      Freq.     Percent        Cum.
------------+----------------------------------- (2)
    January |      1,864       48.98       48.98
      March |      1,942       51.02      100.00
------------+-----------------------------------
      Total |      3,806      100.00
```

1. 3,806 persons = 1,864 + 1,942
2. only January and March

YOUR TURN · TASK 8

Save the quarter

DO

Add and run:

```
save "data/lfs_q1_appended.dta", replace
```

YOU SHOULD SEE

`file data/lfs_q1_appended.dta saved.`

IF NOT

`file already exists?` Add `, replace`.

Part 1 on one page

CHEAT CARD · KEEP THIS ONE

`use "data/jan.dta", clear`	the first month in memory
`append using "f2.dta" "f3.dta", generate(source)`	add the rows of more files
`tabulate source`	which file each row came from
`describe, short`	rows and variables in four lines
`assert _N == 5888`	stop if the row count is wrong
`save "data/q1.dta", replace`	keep the quarter

Practice on your own

PRACTICE SET · practice_D2.do, PART 1

CORE P1 Append February and March only; count the persons.

CORE P2 Same, with `generate(file)`; tabulate `file`.

CORE P3 Assert the result has 4,024 rows.

EXTENSION P4 Cross-tabulate `file` and `int_month`: is every row in its own month?

CHALLENGE P5 Count the households of February with `duplicates report unique_hhd`.

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

You have rebuilt NBS's quarterly file from the monthly files.

lost? open `catchup_D2_S1.do`



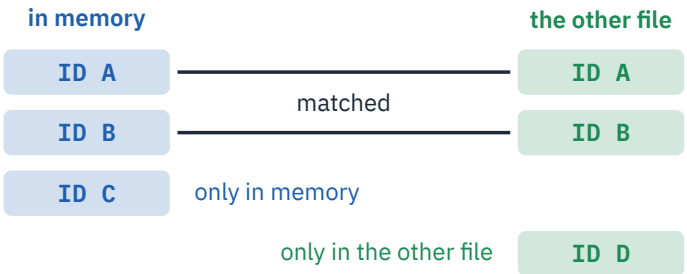
2

DAY 2 · PART 2

Joining files by their key

NBS keeps households, persons and occupation coding in separate files. Merging joins them.

Join rows that carry the same key



YOUR TURN · TASK 9

Look at the household file

DO

Type:

```
use "data/lfs_hh.dta", clear
describe, short
isid unique_hhd
```

YOU SHOULD SEE

1,858 households, and a silent `isid`: one row per household.

IF NOT

Red under `isid`? Tell the trainer: that file must have one row per household.

EXTRA `describe`: which variables describe the household, not the person?

YOUR TURN · TASK 10

Look at the person file

DO

Type:

```
use "data/lfs_person.dta", clear
describe, short
isid unique_ind
```

YOU SHOULD SEE

5,888 persons, and a silent `isid`.

IF NOT

Red under `isid`? Check the variable name: `unique_ind`.

Two files, two levels

HOUSEHOLDS

```
. describe, short

Contains data from data/lfs_hh.dta
 Observations:         1,858
    Variables:             9                  1 Oct 2026 21:10
Sorted by:

. isid unique_hhd
```

1

PERSONS

```
. describe, short

Contains data from data/lfs_person.dta
 Observations:         5,888
    Variables:            77                  6 Oct 2026 19:14
Sorted by: unique_ind

. isid unique_ind
```

2

1,858 households with 9 variables; 5,888 persons with 77. Each `isid` is silent: each file has a unique key.

COMMON ERROR

The household key, tested on persons

```
. isid unique_hhd
variable unique_hhd does not uniquely identify the observations
```

WHY

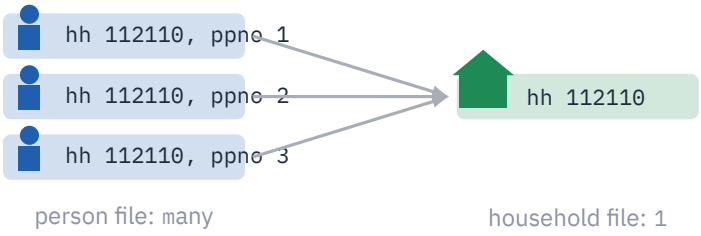
Several persons share one household, so `unique_hhd` repeats in the person file. That is expected.

FIX

In the person file the key is `unique_ind`. The household key is unique only in the household file.

Check the key in both files before every merge.

Many persons, one household



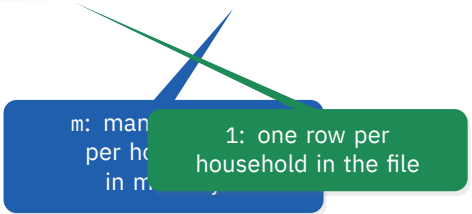
Join the household file onto the persons:

```
merge m:1 unique_hhd using "data/lfs_hh.dta"
```



Read the two sides of the colon

```
merge m:1 unique_hhd using "data/lfs_hh.dta"
```



Left of the colon: the file in memory. Right: the using file.

YOUR TURN · TASK 11

Join households and persons

DO

Add to `my_day2.do` and run:

```
use "data/lfs_person.dta", clear
merge m:1 unique_hhd using "data/lfs_hh.dta"
```

YOU SHOULD SEE

A small report: rows **not matched** and **matched**, and a new variable `_merge`.

IF NOT

`variable unique_hhd does not uniquely identify observations in the using data`? Tell the trainer.

Your result

```
. merge m:1 unique_hhd using "data/lfs_hh.dta"

    Result                           Number of obs
    -----------------------------------------
    Not matched                                 0
    Matched                             5,888  (_merge==3)
    -----------------------------------------
```

Read the report

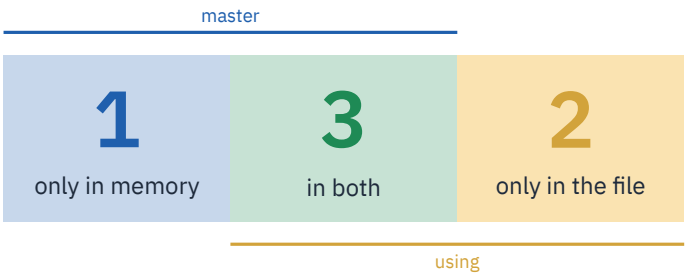
1. **matched**: `_merge` = 3
2. **from master**: `_merge` = 1
3. **from using**: `_merge` = 2

Every row gets one of these values. Here: 0 not matched, all 5,888 matched.

```
. merge m:1 unique_hhd using "data/lfs_hh.dta"

    Result                      Number of obs
    -----------------------------------------
    Not matched                             0
    Matched                             5,888  (_merge==3)
    -----------------------------------------
```

`_merge`, drawn



Make the merge check itself

```
merge m:1 unique_hhd using "data/lfs_hh.dta", assert(match) nogenerate
```

stop if any row is not matched

do not create _merge

The merge either matches every row or stops the do-file.

YOUR TURN · TASK 12

Redo the merge, with its check

DO

Replace the merge line in `my_day2.do` with:

```
merge m:1 unique_hhd using "data/lfs_hh.dta", ///
  assert(match) nogenerate
```

Run the file from the top.

YOU SHOULD SEE

The same report, no red text, and no `_merge` variable.

IF NOT

`option / not allowed?` `///` works only in a do-file: run it from the Do-file Editor.

Your result

```
merge m:1 unique_hhd using "data/lfs_hh.dta", assert(match) nogenerate

    Result                      Number of obs
    -----------------------------------------
    Not matched                             0
    Matched                             5,888
    -----------------------------------------
```

After the merge: the household variables are there

```
. describe, short

Contains data from data/lfs_person.dta
 Observations:         5,888 (1)
    Variables:            85                  6 Oct 2026 19:14
Sorted by: (2)
     Note: Dataset has changed since last saved.
```

(1) still 5,888 persons: the merge added columns, not rows

(2) 85 variables: 77 + the 8 household variables

COMMON ERROR

The wrong colon side

```
. merge 1:1 unique_hhd using "data/lfs_hh.dta"
variable unique_hhd does not uniquely identify observations in the master data
```

WHY

1:1 says “one row per household in memory. The person file has several rows per household.

FIX

Persons in memory, households in the file: `m:1`.

COMMON ERROR

When assert(match) stops a merge

```
. merge m:1 unique_hhd using `h', assert(match)
after merge, not all observations matched
(merged result left in memory)
```

WHY

One household was removed from the file on purpose: its persons found no match, and the check stopped the do-file.

FIX

Find the unmatched persons with `merge` without `assert()` and `tabulate _merge`; then fix the input, not the check.

One person, one coding row:

```
merge 1:1 unique_ind using "data/lfs_coding.dta"
```

one in memory, one in the file

the person key

occupation and industry codes

YOUR TURN · TASK 13

Join the coding file

DO

Add and run:

```
merge 1:1 unique_ind using "data/lfs_coding.dta"
tabulate _merge
```

YOU SHOULD SEE

matched for the employed and **not matched** from master for everyone else.

IF NOT

`variable _merge already defined`? Check `nogenerate` on the household merge.

Your result

```
. merge 1:1 unique_ind using "data/lfs_coding.dta"

    Result                      Number of obs
    -----------------------------------------
    Not matched                         2,876
        from master                     2,876  (_merge==1)
        from using                          0  (_merge==2)

    Matched                             3,012  (_merge==3)
    -----------------------------------------

.
. tabulate _merge

  Matching result from |
                 merge |      Freq.     Percent        Cum.
-----------------------+-----------------------------------
       Master only (1) |      2,876       48.85       48.85
           Matched (3) |      3,012       51.15      100.00
-----------------------+-----------------------------------
                 Total |      5,888      100.00
```

Read the full merge report

```
. merge 1:1 unique_ind using "data/lfs_coding.dta"

    Result                      Number of obs
    -----------------------------------------
    Not matched                         2,876  1
        from master                     2,876  (_merge==1)  2
        from using                          0  (_merge==2)  3

    Matched                             3,012  (_merge==3)  4
    -----------------------------------------
```

1. not matched: 2,876 in all
2. from master: persons with no coding row
3. from using: coding rows with no person: none
4. matched: 3,012, the employed

QUICK CHECK

The report says “from using 0. What does that tell you?

A. the coding file was empty

B. every coding row found its person

C. nobody is employed

ANSWER

The report says “from using 0. What does that tell you?

A the coding file was empty

B every coding row found its person

C nobody is employed

Rows that exist only in the using file would be codes without a person. Zero means the coding team coded only persons in the survey.

COMMON ERROR

Merging twice

```
. merge 1:1 unique_ind using "data/lfs_coding.dta"
variable _merge already defined
```

WHY

The previous merge left `_merge` in memory, and the next merge wants to create it again.

FIX

Use `nogenerate` when you do not need `_merge`, or `drop _merge` after you have read it.

The persons not matched are those without a job. Now you know where the `_merge` in NBS's file came from.

YOUR TURN · TASK 14

Prove it

DO

Type:

```
tabulate _merge ilo_lfs19, missing
```

YOU SHOULD SEE

matched only in the Employed column; every other column in **master only**.

IF NOT

`variable _merge not found`? Task 13 must run without `nogenerate`.

EXTRA Write an `assert` that every matched person is employed.

Your result

```
. tabulate _merge ilo_lfs19, missing
```

Matching result from merge	Labour Force Status - 19th ICLS				
	1 - Emplo	2 - Unemp	3 - Outsi	.	Total
Master only (1)	0	114	1,501	1,261	2,876
Matched (3)	3,012	0	0	0	3,012
Total	3,012	114	1,501	1,261	5,888

Bring only the variables you need

```
merge 1:1 unique_ind using "data/lfs_coding.dta",
keepusing(ilo_job1_ocu_isco08)
```

only these variables come across

Useful when the other file is large.

keepusing() in the output

```
. describe, short

Contains data from data/lfs_person.dta
 Observations:         5,888 (1)
    Variables:            79                  6 Oct 2026 19:14
Sorted by:
     Note: Dataset has changed since last saved.
```

(1) 79 variables: 77 + only the 2 asked for (85 with all of them)

ON YOUR OWN · B

District names on every person

THE QUESTION

NBS keeps district names in `data/district_lookup.dta`. Put `district_name` and `island` on every person, then count persons by island.

HINT

The lookup key is called `district_code`: rename `district_q` to match, then `merge m:1 ... keepusing()`.

ON YOUR OWN · B · ANSWER

```
rename district_q district_code
merge m:1 district_code using
"data/district_lookup.dta", ///
keepusing(district_name island)
tabulate island
```

```
    Result                      Number of obs
    -----------------------------------------
    Not matched                             0  (1)
    Matched                             5,888  (_merge==3)
. tabulate island
     island |      Freq.     Percent        Cum.
------------+-----------------------------------
   La Digue |        120        2.04        2.04  (2)
      Mahé |      5,374       91.27       93.31
    Praslin |        394        6.69      100.00
------------+-----------------------------------
      Total |      5,888      100.00
```

1. all 5,888 matched
2. Mahé 5,374, Praslin 394, La Digue 120

RULE

Never `merge m:m`.

It pairs rows by their position, not by the key. The result looks complete and is wrong.

Find the key that makes one side unique, then use 1:1 or m:1.

YOUR TURN · TASK 15

Final check of the joined file

DO

Add and run:

```
count
isid unique_ind
```

YOU SHOULD SEE

5,888 and a silent `isid`: the same persons as yesterday's clean file.

IF NOT

More than 5,888? A merge added rows: look at `_merge == 2`.

Part 2 on one page

CHEAT CARD · KEEP THIS ONE

`isid key`	is the key unique in this file?
`merge m:1 key using "f.dta"`	many rows in memory, one in the file
`merge 1:1 key using "f.dta"`	one row each side
`..., assert(match)`	stop unless every row matches
`..., nogenerate`	do not create _merge
`..., keepusing(v1 v2)`	bring only these variables
`tabulate _merge`	1 master only, 2 using only, 3 both
`never merge m:m`	find the key that makes one side unique

Practice on your own

PRACTICE SET · practice_D2.do, PART 2

CORE P1 Merge the household file onto the persons, keeping only the weights.

CORE P2 Tabulate `_merge` after joining the coding file. How many matched?

CORE P3 Join the district lookup and tabulate `island` by `ilo_lfs19`.

EXTENSION P4 Prove with `assert` that every matched person is employed.

CHALLENGE P5 Drop one household from a copy of lfs_hh.dta and show which persons no longer match.

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

You have joined three NBS files and proved the result is complete.

lost? open **catchup_D2_S2.do**



3

DAY 2 · PART 3

The same table, turned

Long and wide hold the same numbers

long

district_q	int_month	persons
English River	January	118
English River	February	31
English River	March	155
Mont Buxton	February	158
Saint Louis	February	153

reshape

wide

district_q	persons1	persons2	persons3
English River	118	31	155
Mont Buxton	·	158	·
Saint Louis	·	153	·

1, 2, 3 are the values behind
January, February, March

YOUR TURN · TASK 16

Open the district × month file

DO

Type:

```
use "data/lfs_district_month.dta", clear
list in 1/6
```

YOU SHOULD SEE

One row per district and month: the long layout.

IF NOT

`file not found`? Check the name: `lfs_district_month`.

EXTRA `count`: how many district-months?

Your result

```
. use "data/lfs_district_month.dta", clear

.
. list in 1/6
```

	district_q	int_mo~h	persons
1.	English River	January	118
2.	English River	February	31
3.	English River	March	155
4.	Mont Buxton	February	158
5.	Saint Louis	February	153
6.	Bel Air	January	124

The long file, described

```
. describe

Contains data from data/lfs_district_month.dta
 Observations:            47                  1
    Variables:             3                  1 Oct 2026 21:10
-------------------------------------------------------------------------------
Variable      Storage   Display    Value
    name         type    format    label      Variable label
-------------------------------------------------------------------------------
district_q      byte    %26.0g     DISTRICT_Q
                                              DISTRICT
int_month       byte    %8.0g      int_month
                                              Interview month, Q1 2026
persons         int     %12.0g                2
-------------------------------------------------------------------------------
Sorted by: district_q  int_month              3
```

1. 47 rows: one per district and month with interviews
2. `persons`: the count in each cell
3. sorted by district, then month: the long layout

Turn long into wide:



Rows from `i()`, columns from `j()`

```
reshape wide persons, i(district_q) j(int_month)
```



YOUR TURN · TASK 17

Reshape to wide

DO

Type:

```
reshape wide persons, i(district_q) j(int_month)
list in 1/6
```

YOU SHOULD SEE

26 rows, with `persons1`, `persons2`, `persons3`; some cells empty.

IF NOT

`variable persons not found`? Check the name in the Variables window.

Your result

```
. reshape wide persons, i(district_q) j(int_month)
(j = 1 2 3)

Data                                Long   ->   Wide
-----------------------------------------------------------------------------
Number of observations                47   ->   26
Number of variables                    3   ->   4
j variable (3 values)          int_month   ->   (dropped)
xij variables:
                                 persons   ->   persons1 persons2 persons3
-----------------------------------------------------------------------------

.
. list in 1/6

     +-------------------------------------------------+
     | district_q   persons1   persons2   persons3 |
     |-------------------------------------------------|
  1. | English River       118         31       155 |
  2. |   Mont Buxton         .        158         . |
  3. |   Saint Louis         .        153         . |
  4. |       Bel Air       124         10       132 |
  5. |   Mont Fleuri         .        157         . |
     |-------------------------------------------------|
  6. |     Plaisance       135          5       124 |
     +-------------------------------------------------+
```

Read the reshape summary

```
. reshape wide persons, i(district_q) j(int_month)
(j = 1 2 3)

Data                                Long   ->   Wide
-----------------------------------------------------------------------------
Number of observations                47   ->   26
Number of variables                    3   ->   4
j variable (3 values)          int_month   ->   (dropped)
xij variables:
                                 persons   ->   persons1 persons2 persons3
-----------------------------------------------------------------------------
```

1. 47 long rows become 26 wide rows: one per district
2. 3 variables become 4
3. `persons` becomes `persons1`, `persons2`, `persons3`

An empty cell here means zero

```
replace persons2 = 0 if missing(persons2)
```

not visited that month:
0 persons interviewed

Only because the cell counts people. A missing income stays missing.

A total across months: two ways

```
. generate total = persons1 + persons2 + persons3
(18 missing values generated)

. egen total2 = rowtotal(persons1 persons2 persons3)

. list in 1/6

     +-------------------------------------------------------------------+
     |    district_q   persons1   persons2   persons3   total   total2 |
     |-------------------------------------------------------------------|
  1. | English River        118         31        155     304       304 |
  2. |   Mont Buxton          .        158          .       .       158 |
  3. |   Saint Louis          .        153          .       .       153 |
  4. |       Bel Air        124         10        132     266       266 |
  5. |   Mont Fleuri          .        157          .       .       157 |
     |-------------------------------------------------------------------|
  6. |     Plaisance        135          5        124     264       264 |
     +-------------------------------------------------------------------+
```

1. \+ gives missing when any month is missing: 18 districts lose their total
2. egen `rowtotal()` treats missing as 0: Mont Buxton 158

Empty months set to zero

```
. replace persons1 = 0 if missing(persons1)
(13 real changes made) 1

. replace persons2 = 0 if missing(persons2)
(5 real changes made) 2

. replace persons3 = 0 if missing(persons3)
(13 real changes made) 3
```

1. 13 districts had no January interviews
2. 5 had none in February
3. 13 had none in March

COMMON ERROR

reshape without i()

```
. reshape wide persons, j(int_month)
option i() required
```

WHY

Stata needs to know which variable identifies a row in the wide file. It then prints a reminder of the syntax.

FIX

```
reshape wide persons, i(district_q)
j(int_month)
```

And back to long:

```
reshape long persons i(district_q) j(int_month)
```



YOUR TURN · TASK 18

Back to long

DO

Type:

```
reshape long persons, i(district_q) j(int_month)
count
```

YOU SHOULD SEE

78 rows: 26 districts × 3 months, now including the empty months.

IF NOT

`variable persons not found`? Run Task 17 first: the file must be wide.

EXTRA Type `reshape wide` alone: Stata repeats the last reshape.

Back to long: 78 rows

```
. reshape long persons, i(district_q) j(int_month)
(j = 1 2 3)

Data                               Wide   ->   Long
-----------------------------------------------------------------------------
Number of observations               26   ->   78
Number of variables                   4   ->   3
j variable (3 values)                     ->   int_month
xij variables:
             persons1 persons2 persons3   ->   persons
-----------------------------------------------------------------------------
. list in 1/6

     +-----------------------------------+
     |    district_q   int_mo~h   persons |
     |-----------------------------------|
  1. | English River    January       118 |
  2. | English River   February        31 |
  3. | English River      March       155 |
  4. |   Mont Buxton    January         . |
  5. |   Mont Buxton   February       158 |
     |-----------------------------------|
  6. |   Mont Buxton      March         . |
     +-----------------------------------+
```

1. 26 wide rows become 78 long rows
2. Mont Buxton, January: a row with no persons

QUICK CHECK

You started with 47 rows. After reshape wide and back to long you have 78. Why?

A. reshape duplicated some rows

B. every district now has a row for every month, empty or not

C. the months were counted twice

ANSWER

You started with 47 rows. After reshape wide and back to long you have 78. Why?

A reshape duplicated some rows

B every district now has a row for every month, empty or not

C the months were counted twice

The wide file has a column for every month, so the long file gets 26 x 3 = 78 rows. The 31 new rows are the empty district-months.

ON YOUR OWN · C

The CPI arrives wide

THE QUESTION

The CPI file `cpi_divisions_wide.dta` has one row per month and `cpi1` to `cpi12` for the COICOP divisions. Turn it into one row per month and division.

HINT

`reshape long`: the stub is `cpi`, the row is `month`, the new `j()` is `division`.

ON YOUR OWN · C · ANSWER

```
use "data/cpi_divisions_wide.dta", clear
reshape long cpi, i(month) j(division)
list in 1/3
```

```
. reshape long cpi, i(month) j(division)
Data                               Wide   ->   Long
-----------------------------------------------------------------------------
Number of observations              236   ->   2,832
Number of variables                  13   ->   3
j variable (12 values)                    ->   division
xij variables:
                     cpi1 cpi2 ... cpi12   ->   cpi
-----------------------------------------------------------------------------
. list in 1/3
     +-------------------------------------------------------+
     |  month                           division         cpi |
     |-------------------------------------------------------|
  1. | 2007m1   Food and non-alcoholic beverages   41.768899 |
  2. | 2007m1    Alcoholic beverages and tobacco   45.865009 |
  3. | 2007m1              Clothing and footwear    41.26573 |
     +-------------------------------------------------------+
```

1. 236 months × 12 divisions = 2,832 rows
2. `division` takes the numbers 1 to 12 from `cpi1` … `cpi12`
3. one row per month and division

Part 3 on one page

CHEAT CARD · KEEP THIS ONE

`reshape wide v, i(row) j(col)`	long to wide: values of col become suffixes
`reshape long v, i(row) j(col)`	wide to long: suffixes become values
`egen t = rowtotal(v1 v2 v3)`	a total across columns, missing as 0
`replace v1 = 0 if missing(v1)`	only when empty means zero

Practice on your own

PRACTICE SET · practice_D2.do, PART 3

CORE	P1 Make `lfs_district_month.dta` wide. How many districts had no February interview?
CORE	P2 Add `total`, the persons of the quarter in each district, with `egen rowtotal()`.
CORE	P3 Assert that the 26 totals add up to 5,888.
EXTENSION	P4 Share of each district's persons interviewed in March, one decimal.
CHALLENGE	P5 Turn the CPI wide file long and count the rows.

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

Release tables are usually wide; analysis files are usually long.

You can turn any NBS table between its analysis and publication shapes.

lost? open `catchup_D2_S3.do`



4

DAY 2 · PART 4

The tables of the release

MONDAY MORNING AT NBS

The Director needs three numbers for the Q1 note

QUESTION 1

How many persons did we interview in each region?

QUESTION 2

What share of working-age persons is employed, region by region?

QUESTION 3

How many hours a week do employed persons usually work?

By the end of this part you answer all three from one do-file, and you can rebuild them any time.

What you will be able to do

AFTER THIS PART YOU CAN

- count persons by one or two variables, and check the base of every table
- choose row or column percentages for the question asked
- give the mean, the median and the spread of hours by group
- build the same tables with `table`, ready for Excel and Word

YOUR TURN · TASK 19

Open yesterday’s product

DO

Add to `my_day2.do` and run:

```
use "data/lfs_teach.dta", clear
count
```

YOU SHOULD SEE

5,888: the clean analysis file you saved at the end of Day 1.

IF NOT

`file data/lfs_teach.dta not found`? Run `day1/catchup/catchup_D1_S9.do` once, then try again.

Yesterday’s names, today’s tables

NBS NAME (RAW FILE)	TEACHING NAME (lfs_teach.dta)
region_q	**region**
district_q	**district**
psu_nbs_2026	**psu**
ea_q	**ea**
weights_dis	**wgt_final**

The variables you made yesterday are here too: employed, unemployed, agegrp, hhsize.

QUESTION 1

How many persons did we interview in each region?

You know the command from yesterday: `tabulate` with one variable.
Today we read every column of its output.

YOUR TURN · TASK 20

How many persons in each region?

DO

Type:

```
tabulate region
```

YOU SHOULD SEE

Seven regions and a total of 5,888.

IF NOT

`variable region not found`? You are on an older file: run Task 19 again.

EXTRA Which region holds the most records?

Read every column

```
. tabulate region
```

Region	Freq.	Percent	Cum.
1- North	940	15.96	15.96
2- Central 1	881	14.96	30.93
3- Central 2	842	14.30	45.23
4- EastSouth	1,449	24.61	69.84
5- West	900	15.29	85.12
6- Praslin & LaDigue	514	8.73	93.85
7- Perseverance	362	6.15	100.00
Total	5,888	100.00	

1. one row per value of `region`, shown with its label
2. **Freq.**: how many records
3. **Percent**: Freq. as a share of all records
4. **Cum.**: the Percent column added down the rows
5. **Total**: every record in the file, 5,888

QUICK CHECK

In the Cum. column, 3- Central 2 shows 45.23. What does it say?

A. Central 2 holds 45.23% of the persons

B. North, Central 1 and Central 2 together hold 45.23% of the persons

C. 45.23 persons are interviewed in Central 2 per 100 households

ANSWER

In the Cum. column, 3- Central 2 shows 45.23. What does it say?

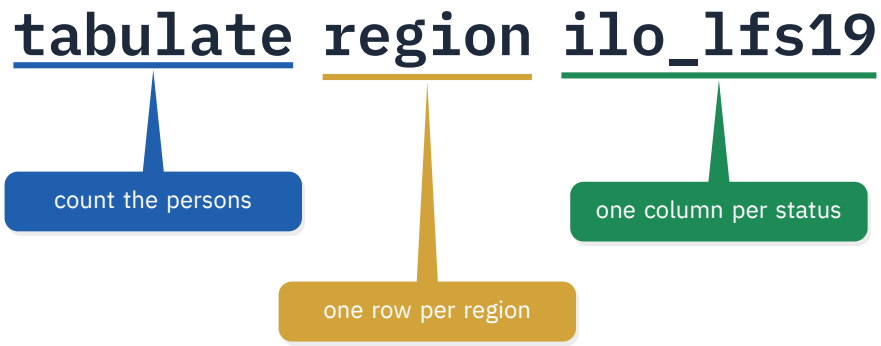
A Central 2 holds 45.23% of the persons

B North, Central 1 and Central 2 together hold 45.23% of the persons

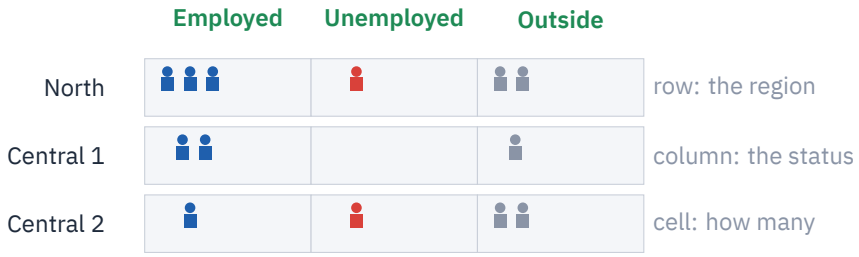
C 45.23 persons are interviewed in Central 2 per 100 households

Cum. adds the Percent column from the top down to that row: (940 + 881 + 842) / 5,888 = 45.23%. Central 2 alone is 14.30%.

Two variables, one table:



Every person lands in exactly one cell



YOUR TURN · TASK 21

Region by labour force status

DO

Type:

```
tabulate region ilo_lfs19
```

YOU SHOULD SEE

Seven rows, three status columns, and a total of 4,627.

IF NOT

`variable ilo_lfs19 not found`? Check the spelling: `lfs`, then `19`.

EXTRA Why 4,627 and not 5,888?

Read a two-way table

```
. tabulate region ilo_lfs19
```

Region	Labour Force Status - 19th ICLS (2) 1 - Emplo	2 - Unemp	3 - Outsi	Total
(1) 1- North	508 (3)	17	203	(4) 728
2- Central 1	462	25	227	714
3- Central 2	383	16	274	673
4- EastSouth	742	23	377	1,142
5- West	466	26	213	705
6- Praslin & LaDigue	277	3	121	401
7- Perseverance	174	4	86	264
Total	3,012 (5)	114	1,501	(6) 4,627

1. rows: the first variable, `region`
2. columns: the second, `ilo_lfs19`; labels cut to fit
3. 508 persons in North are employed
4. row total: 728 persons in North
5. column total: 3,012 employed
6. grand total: 4,627

QUICK CHECK

North has 940 persons (Task 20), but its row in Task 21 adds to 728. Where are the other 212?

A. They were dropped as duplicates

B. They are under 15: never asked the labour questions, so their status is missing

C. They are in the Total row

ANSWER

North has 940 persons (Task 20), but its row in Task 21 adds to 728. Where are the other 212?

A They were dropped as duplicates

B They are under 15: never asked the labour questions, so their status is missing

C They are in the Total row

`ilo_lfs19` is missing for everyone under 15, and `tabulate` leaves missing values out without saying so.

CHECK THE BASE

`tabulate` leaves out missing values.

Task 21 counts 4,627 persons, not 5,888: the 1,261 under-15s have no labour force status.

Add `, missing` once, to see the base of every table.

YOUR TURN · TASK 22

Show the missing column

DO

Type:

```
tabulate region ilo_lfs19, missing
```

YOU SHOULD SEE

A fourth column headed `.` and a total of 5,888.

IF NOT

Total still 4,627? The `, missing` part did not run: check the comma.

The base, made visible

```
. tabulate region ilo_lfs19, missing

                    |       Labour Force Status - 19th ICLS
             Region | 1 - Emplo  2 - Unemp  3 - Outsi          . |     Total
--------------------+--------------------------------------------+----------
          1- North |       508         17        203        212 |       940
    2- Central 1 |       462         25        227        167 |       881
    3- Central 2 |       383         16        274        169 |       842
    4- EastSouth |       742         23        377        307 |     1,449
         5- West |       466         26        213        195 |       900
6- Praslin & LaDigue |       277          3        121        113 |       514
     7- Perseverance |       174          4         86         98 |       362
--------------------+--------------------------------------------+----------
               Total |     3,012        114      1,501      1,261 |     5,888
```

1. the column .: no status recorded
2. 212 persons in North, all under 15
3. 1,261 missing; the grand total is now all 5,888

A short name that still works

```
. tabulate region ilo_lfs
```

Region	Labour Force Status - 19th ICLS 1 - Emplo	2 - Unemp	3 - Outsi	Total
1- North	508	17	203	728
2- Central 1	462	25	227	714
3- Central 2	383	16	274	673
4- EastSouth	742	23	377	1,142
5- West	466	26	213	705
6- Praslin & LaDigue	277	3	121	401
7- Perseverance	174	4	86	264
Total	3,012	114	1,501	4,627

1. typed `ilo_lfs`, not `ilo_lfs19`: Stata still ran the full table

Stata accepts the start of a name when only one variable begins that way. In a do-file, type the full name: another file may also have `ilo_lfs20`.

ON YOUR OWN · D

Women aged 15 to 24

THE QUESTION

How many women aged 15–24 are in the sample?

HINT

Two variables you built or met yesterday: `agegrp` and `ilo_sex`. One table.

ON YOUR OWN · D · ANSWER

```
tabulate agegrp ilo_sex
```

```
           |          Sex
 Age group |  1 - Male  2 - Femal |     Total
-----------+----------------------+----------
     15-24 |       354        366 |       720
     25-34 |       396        393 |       789
     35-54 |       782        873 |     1,655
       55+ |       616        847 |     1,463
-----------+----------------------+----------
     Total |     2,148      2,479 |     4,627
```

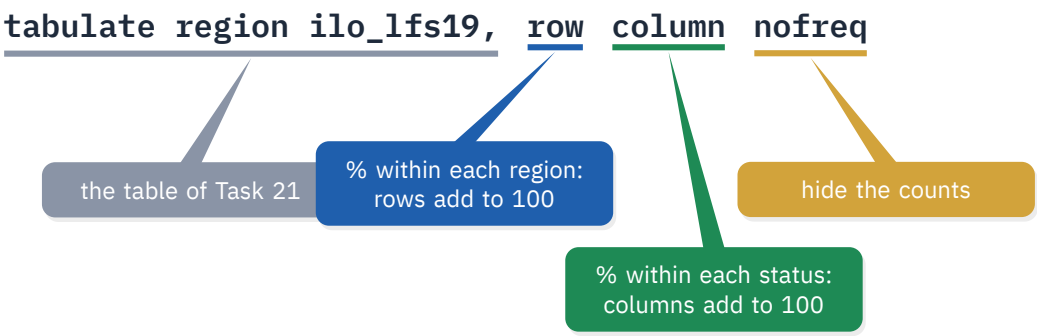
1. 366 women aged 15–24
2. 4,627 in all: `agegrp` is empty under 15, like the status

QUESTION 2

What share of working-age persons is employed, region by region?

Counts cannot answer it: EastSouth has more of everyone. We need the share *within* each region.

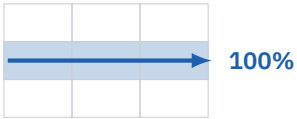
Which percentage?



Use `row` or `column`, one at a time.

The same table, read two ways

`row`



Of the people in this region, what share is employed?

`column`



Of all the employed, what share lives in this region?

YOUR TURN · TASK 23

Status within each region

DO

Type:

```
tabulate region ilo_lfs19, row nofreq
```

YOU SHOULD SEE

Percentages only; every row adds to 100.

IF NOT

`variable row not found`? Put a comma before `row`.

Read row percentages

```
. tabulate region ilo_lfs19, row nofreq
```

Region	Labour Force Status - 19th ICLS 1 - Emplo	2 - Unemp	3 - Outsi	Total
1- North	69.78	2.34	27.88	100.00
2- Central 1	64.71	3.50	31.79	100.00
3- Central 2	56.91	2.38	40.71	100.00
4- EastSouth	64.97	2.01	33.01	100.00
5- West	66.10	3.69	30.21	100.00
6- Praslin & LaDigue	69.08	0.75	30.17	100.00
7- Perseverance	65.91	1.52	32.58	100.00
Total	65.10	2.46	32.44	100.00

1. each row adds to 100
2. of the working-age persons interviewed in North, 69.78% are employed
3. lowest: Central 2, 56.91%
4. all regions together: 65.10%

YOUR TURN · TASK 24

Now the other direction

DO

Type:

```
tabulate region ilo_lfs19, column
nofreq
```

YOU SHOULD SEE

Every **column** adds to 100.

IF NOT

Rows add to 100? You typed `row`: change it to `column`.

Read column percentages

Region	Labour Force Status - 19th ICLS 1 - Emplo	2 - Unemp	3 - Outsi	Total	
1- North	16.87	14.91	13.52	15.73	
2- Central 1	15.34	21.93	15.12	15.43	
3- Central 2	12.72	14.04	18.25	14.55	
4- EastSouth	24.63	20.18	25.12	**24.68**	3
5- West	15.47	**22.81**	14.19	15.24	2
6- Praslin & LaDigue	9.20	2.63	8.06	8.67	
7- Perseverance	5.78	3.51	5.73	5.71	
Total	100.00	100.00	100.00	100.00	1

1. each column adds to 100
2. of all unemployed in the sample, 22.81% live in West
3. last column: each region's share of the working-age sample

QUICK CHECK

A draft press note says: “West holds 22.8% of the unemployed.” Which table gives this number?

A. the row percentages of Task 23

B. the column percentages of Task 24

C. the counts of Task 21

ANSWER

A draft press note says: “West holds 22.8% of the unemployed.” Which table gives this number?

A the row percentages of Task 23

B the column percentages of Task 24

C the counts of Task 21

“Of all unemployed” is the 100%: that is a column of the table. The row table would say what share of West is unemployed (3.69%).

COMMON ERROR

An option without its comma

```
. tabulate region ilo_lfs19 row
variable row not found
```

WHY

Without the comma, Stata reads `row` as a third variable name. There is no variable called `row`.

FIX

Everything after the comma is an option: `tabulate region ilo_lfs19, row`

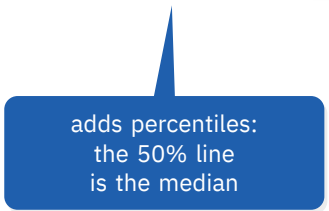
QUESTION 3

How many hours a week do employed persons usually work?

One number is not enough. The note needs the middle, the typical range, and how the groups differ.

The median and the percentiles

```
summarize ilo_job1_how_usual, detail
```



Hours are skewed: report the median next to the mean.

YOUR TURN · TASK 25

The full picture of usual hours

DO

Type:

```
summarize ilo_job1_how_usual, detail
```

YOU SHOULD SEE

Percentiles on the left, the four smallest and largest values in the middle, mean and spread on the right.

IF NOT

Only one line of output? The `, detail` part is missing.

Read summarize, detail

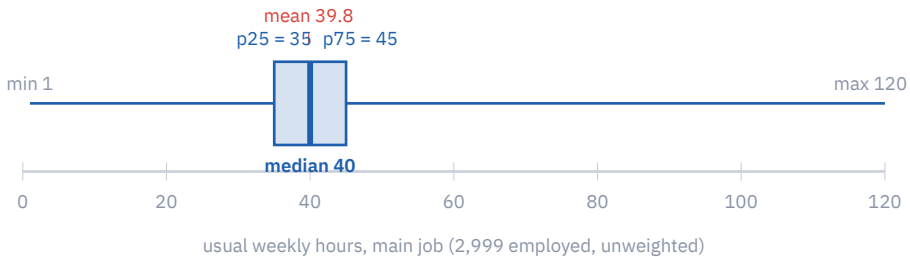
```
. summarize ilo_job1_how_usual, detail

          Weekly hours usually worked - main job
-------------------------------------------------------------
      Percentiles      Smallest
 1%             7              1
 5%            20              2
10%            35              2       Obs                2,999
25%            35              2       Sum of wgt.        2,999

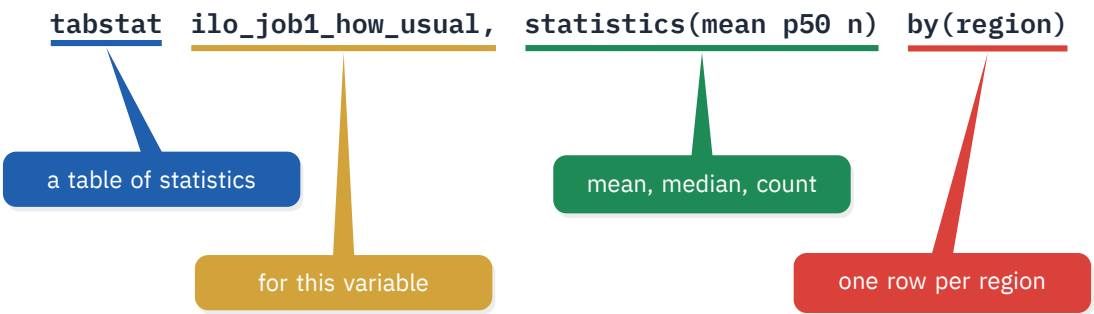
50%            40                      Mean           39.82861
                      Largest          Std. dev.      11.48297
75%            45            105
90%            50            108       Variance       131.8585
95%            60            119       Skewness       .6350666
99%            77            120       Kurtosis       8.582522
```

1. 50%: the median, 40 hours; on the same line the mean, 39.8
2. Obs: 2,999 answers; 13 employed gave none
3. the four largest values: 105 to 120 hours, worth a check with the field team

Half of the employed work 35 to 45 hours



Statistics by group, in one table:



YOUR TURN · TASK 26

Usual hours by region

DO

Add to `my_day2.do` and run:

```
tabstat ilo_job1_how_usual, ///
    statistics(mean p50 n) by(region)
```

YOU SHOULD SEE

One row per region, then Total: the mean, the median and the number of answers.

IF NOT

`option / not allowed?` `///` works only in a do-file.

Read tabstat

```
. tabstat ilo_job1_how_usual, statistics(mean p50 n) by(region)

Summary for variables: ilo_job1_how_usual
Group variable: region (Region)
```

region	Mean	p50	N
1- North	39.88976	35	508
2- Central 1	41.09348	40	460
3- Central 2	39.15223	37	381
4- EastSouth	39.90946	40	740
5- West	39.45474	39	464
6- Praslin & LaD	38.01091	40	275
7- Perseverance	41.33918	40	171
Total	39.82861	40	2999

1. **Mean** usual hours in each region
2. **p50**: the median
3. **N**: answers behind each number
4. Total: all regions together, the same as `summarize`
5. Praslin & LaD: long labels are cut

The statistics tabstat knows

`mean`	the average	`sd`	standard deviation
`p50`	the median	`n`	number of answers
`min, max`	smallest, largest	`sum`	the total
`p25, p75`	the middle half	`p10, p90`	tails

List any of them inside `statistics( )`, in the order you want the columns.

YOUR TURN · TASK 27

The same, by sex

DO

Type:

```
tabstat ilo_job1_how_usual, ///
    statistics(mean p50 n) by(ilo_sex)
```

YOU SHOULD SEE

Two rows, Male and Female, then Total.

IF NOT

`option / not allowed`? Run it from the Do-file Editor.

Men and women

```
. tabstat ilo_job1_how_usual, statistics(mean p50 n) by(ilo_sex)

Summary for variables: ilo_job1_how_usual
Group variable: ilo_sex (Sex)

   ilo_sex |      Mean       p50         N
-----------+--------------------------------
  1 - Male |   41.3522        40      1431
2 - Female |  38.43814        35      1568
-----------+--------------------------------
     Total |  39.82861        40      2999
----------------------------------------------
```

1. men: mean 41.4, median 40
2. women: mean 38.4, median 35: the middle employed woman works 35 hours

YOUR TURN · TASK 28

Add the extremes

DO

Change the line to:

```
statistics(mean p50 n min max)
```

YOU SHOULD SEE

Two more columns, Min and Max.

IF NOT

Columns in another order? They follow the order inside `statistics( )`.

Min and max

```
. tabstat ilo_job1_how_usual, statistics(mean p50 n min max) by(ilo_sex)

Summary for variables: ilo_job1_how_usual
Group variable: ilo_sex (Sex)

   ilo_sex |      Mean       p50         N       Min       Max
-----------+--------------------------------------------------
  1 - Male |   41.3522        40      1431         1       120
2 - Female |  38.43814        35      1568         2       108
-----------+--------------------------------------------------
     Total |  39.82861        40      2999         1       120
----------------------------------------------------------------
```

1. a man reports 120 hours: check before publishing
2. women: 2 to 108 hours

ON YOUR OWN · E

The spread of hours, region by region

THE QUESTION

For each region: minimum, 25th percentile, median, 75th percentile and maximum of usual hours. Which region has the highest 75th percentile?

HINT

One `tabstat`; five names inside `statistics( )`.

ON YOUR OWN · E · ANSWER

```
tabstat ilo_job1_how_usual, ///
statistics(min p25 p50 p75 max) by(region)
```

region	Min	p25	p50	p75	Max
1- North	7	35	35	44	120
2- Central 1	1	35	40	45	108
3- Central 2	2	35	37	45	98
4- EastSouth	2	35	40	45	90
5- West	2	35	39	45	119
6- Praslin & LaD	3	35	40	48	105
7- Perseverance	14	35	40	45	84
Total	1	35	40	45	120

1

1 Praslin & La Digue: a quarter of its employed work 48 hours or more

QUICK CHECK

For women, the mean is 38.4 hours and the median 35. What does that tell you?

A. most women work exactly 38.4 hours

B. some long working weeks pull the mean above the middle

C. the data contain errors

ANSWER

For women, the mean is 38.4 hours and the median 35. What does that tell you?

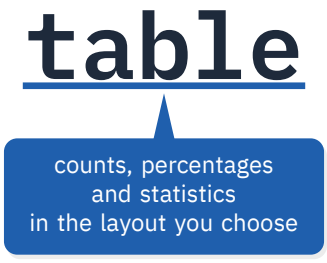
A most women work exactly 38.4 hours

B some long working weeks pull the mean above the middle

C the data contain errors

The median is the middle person. A tail of long weeks (up to 108) raises the mean but not the middle. That is why release tables often give both.

One command for the release tables:



Stata 17 or newer.

Three commands, one job each

tabulate

counts and percentages of one or two variables

quick look, checks

tabstat

statistics of a variable, by one group

quick summaries

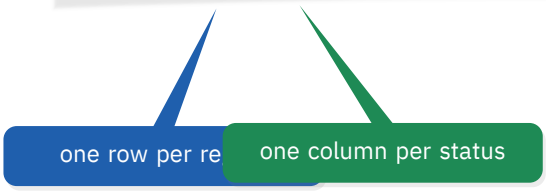
table

counts, percentages and statistics, any layout

release tables: goes straight to Excel and Word

Rows first, then columns

```
table (region) (ilo_lfs19)
```



With no statistic named, `table` counts persons.

YOUR TURN · TASK 29

Region by status, with `table`

DO

Type:

```
table (region) (ilo_lfs19)
```

YOU SHOULD SEE

The counts of Task 21, now with a Total row and column.

IF NOT

A table in a different shape? Check the two pairs of brackets.

Read table

```
. table (region) (ilo_lfs19)
```

	Labour Force Status - 19th ICLS			
	1 - Employed	2 - Unemployed	3 - Outside Labour Force	Total
Region				
1- North	508	17	203	728
2- Central 1	462	25	227	714
3- Central 2	383	16	274	673
4- EastSouth	742	23	377	1,142
5- West	466	26	213	705
6- Praslin & LaDigue	277	3	121	401
7- Perseverance	174	4	86	264
Total	3,012	114	1,501	4,627

1. full status labels, not cut
2. the row variable is named once, as a heading
3. Total column
4. Total row: same as Task 21

Row percentages in `table`

```
table (region) (ilo_lfs19), statistic(percent, across(ilo_lfs19))
```

percentages, not counts

adding to 100 across the statuses: row %

The same shares as `tabulate …, row`.

YOUR TURN · TASK 30

Row percentages, one decimal

DO

Add to `my_day2.do` and run:

```
table (region) (ilo_lfs19), ///
    statistic(percent, across(ilo_lfs19)) ///
    nformat(%5.1f)
```

YOU SHOULD SEE

The row percentages of Task 23, one decimal, in the `table` layout.

IF NOT

Red text? Count the brackets: `across(ilo_lfs19))` ends with two.

Read the release-ready table

	Labour Force Status - 19th ICLS			
	1 - Employed	2 - Unemployed	3 - Outside Labour Force	Total
Region				
1- North	69.8	2.3	27.9	100.0
2- Central 1	64.7	3.5	31.8	100.0
3- Central 2	56.9	2.4	40.7	100.0
4- EastSouth	65.0	2.0	33.0	100.0
5- West	66.1	3.7	30.2	100.0
6- Praslin & LaDigue	69.1	0.7	30.2	100.0
7- Perseverance	65.9	1.5	32.6	100.0
Total	65.1	2.5	32.4	100.0

1. North: 69.8% employed; the row adds to 100.0

2. Total: 65.1%, the same as Task 23

`across()` chooses the direction

```
statistic(percent, across(ilo_lfs19))
statistic(percent, across(region))
```

- adds to 100 across the statuses: **row** %, status within region
- adds to 100 across the regions: **column** %, region within status

YOUR TURN · TASK 31

Column percentages with `table`

DO

Change `across()` and run:

```
table (region) (ilo_lfs19), ///
    statistic(percent, across(region)) ///
    nformat(%5.1f)
```

YOU SHOULD SEE

Every column adds to 100.0, as in Task 24.

IF NOT

Rows add to 100.0? `across()` still names `ilo_lfs19`.

Column percentages, release-ready

	Labour Force Status - 19th ICLS			
	1 - Employed	2 - Unemployed	3 - Outside Labour Force	Total
Region				
1- North	16.9	14.9	13.5	15.7
2- Central 1	15.3	21.9	15.1	15.4
3- Central 2	12.7	14.0	18.3	14.5
4- EastSouth	24.6	20.2	25.1	24.7
5- West	15.5	**22.8**	14.2	15.2
6- Praslin & LaDigue	9.2	2.6	8.1	8.7
7- Perseverance	5.8	3.5	5.7	5.7
Total	100.0	100.0	100.0	100.0

1. West: 22.8% of the unemployed (Task 24: 22.81)
2. each column adds to 100.0

COMMON ERROR

One bracket short

```
. table (region) (ilo_lfs19), statistic(percent, across(ilo_lfs19)
) required
```

WHY

`statistic(` opens a bracket and `across(` opens another. Only one is closed.

FIX

Close both: `across(ilo_lfs19))`. In the Do-file Editor, click next to a bracket: its partner lights up.

Two statistics in each cell

```
table (region) (ilo_sex), ///
    statistic(mean ilo_job1_how_usual) ///
    statistic(count ilo_job1_how_usual)
```

- the mean usual hours
- how many answers that mean is based on

Always print the count next to a mean.

YOUR TURN · TASK 32

Mean hours and counts, by region and sex

DO

Add to `my_day2.do` and run:

```
table (region) (ilo_sex), ///
    statistic(mean ilo_job1_how_usual) ///
    statistic(count ilo_job1_how_usual) ///
    nformat(%4.1f mean)
```

YOU SHOULD SEE

Under each region two rows, **Mean** and **Number of nonmissing values**.

IF NOT

`option / not allowed`? Run from the Do-file Editor.

Read two statistics per cell

	Sex			
	1 - Male	2 - Female	Total	
Region				
1- North				
Mean	**41.3**	**38.5**	39.9	1
Number of nonmissing values	249	259	508	2
Total				
Mean	41.4	38.4	39.8	3
Number of nonmissing values	1,431	1,568	2,999	

1. North: men 41.3, women 38.5 hours
2. behind them: 249 and 259 answers
3. all regions: 41.4 and 38.4

ON YOUR OWN · F

Who is unemployed?

THE QUESTION

Among working-age persons, which age group has the highest share unemployed? Use `table`, row percentages, one decimal.

HINT

Rows: `agegrp`. Columns: `ilo_lfs19`. Then `statistic(percent, across(...))`.

ON YOUR OWN · F · ANSWER

```
table (agegrp) (ilo_lfs19), ///
statistic(percent, across(ilo_lfs19)) ///
nformat(%5.1f)
```

	Labour Force Status - 19th ICLS			
Age group	1 - Employed	2 - Unemployed	3 - Outside Labour Force	Total
15-24	42.4	7.8	49.9	100.0
25-34	84.8	2.8	12.4	100.0
35-54	85.7	1.7	12.6	100.0
55+	42.4	0.5	57.1	100.0
Total	65.1	2.5	32.4	100.0

1

1 15–24: 7.8% unemployed, three times the total (2.5%)

QUICK CHECK

You need the share of each region among the employed. Which option?

A `statistic(percent, across(ilo_lfs19))`

B `statistic(percent, across(region))`

C `nformat(%5.1f)`

ANSWER

You need the share of each region among the employed. Which option?

A `statistic(percent, across(ilo_lfs19))`

B `statistic(percent, across(region))`

C `nformat(%5.1f)`

“Among the employed” makes the employed column the 100%: the percentages add up across the regions.

STATA 18 OR NEWER

`dtable`: a “Table 1” of characteristics in one line

```
. dtable ilo_age ilo_job1_how_usual, by(ilo_sex)

------------------------------------------------------------------------------------------
                                                                Sex
                                            1 - Male       2 - Female        Total
------------------------------------------------------------------------------------------
N                                         2,796 (47.5%)   3,092 (52.5%)  5,888 (100.0%)
Age in completed years (top-coded at 85) 35.058 (21.510) 38.175 (22.255) 36.695 (21.958)
Weekly hours usually worked - main job   41.352 (13.000)  38.438 (9.695) 39.829 (11.483)
------------------------------------------------------------------------------------------
```

NBS runs 19.5, so it works in the office. On Stata 17, build the same table with `table`.

Part 4 on one page

CHEAT CARD · KEEP THIS ONE

`tabulate v`	counts and shares of one variable
`tabulate r c`	two-way counts
`tabulate r c, missing`	show the missing as a column
`tabulate r c, row nofreq`	shares within each row
`tabulate r c, column nofreq`	shares within each column
`summarize v, detail`	percentiles, mean, spread, extremes
`tabstat v, statistics(mean p50 n) by(g)`	statistics by group
`table (r) (c)`	release table of counts
`table (r) (c), statistic(percent, across(c))`	row percentages
`table ..., statistic(mean v) statistic(count v)`	mean with its count
`..., nformat(%5.1f)`	one decimal

Practice on your own

PRACTICE SET · practice_D2.do, PART 4

CORE P1 Count persons by marital status (`dem_mrt`), showing the missing.

CORE P2 Share of women and men in each age group (row %, one decimal, with `table`).

CORE P3 Median usual hours by age group.

EXTENSION P4 Mean and count of usual hours by region and age group, mean with one decimal.

CHALLENGE P5 Which district has the highest share employed among working-age persons? (Hint: rows `district`.)

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

You can produce the counts, shares and averages of any release table.

lost? open **catchup_D2_S4.do**

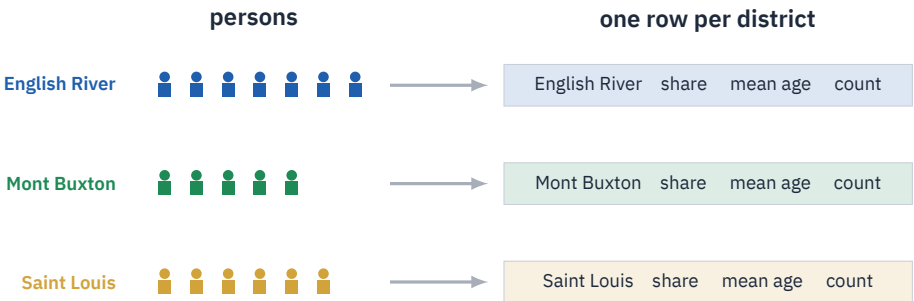


5

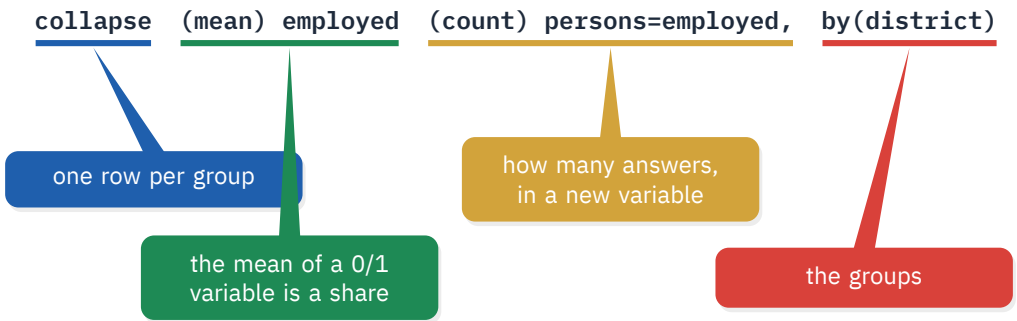
DAY 2 · PART 5

From persons to districts

Person rows become district rows



Statistics by group, kept as data:



`collapse` replaces the data in memory.

The 5,888 persons are gone until you open the file again.

Protect them first: `preserve`.

A photo of the data, then back to it

```
preserve
collapse …
list
restore
```

- `preserve`: a photo of the 5,888 persons
- `restore`: put them back

Run the whole block in one go: the photo lasts until the run ends.

YOUR TURN · TASK 33

One row per district

DO

Add to `my_day2.do`, then run the block:

```
preserve
collapse (mean) employed ilo_age ///
  (count) persons=employed ///
  if !missing(employed), by(district)
list in 1/5
```

YOU SHOULD SEE

26 district rows; `employed` is now a share, `persons` counts people aged 15 and over.

IF NOT

`( is not a valid command name`? The lines ran one by one: run the block from the Do-file Editor.

EXTRA Add `egen rank = rank(-employed)`: which district ranks 1?

Your result

```
. collapse (mean) employed ilo_age (count) persons=employed if !missing(employed), by(district)

.
. list in 1/5
```

	district	employed	ilo_age	persons
1.	English River	.6598361	42.97541	244
2.	Mont Buxton	.648	45.16	125
3.	Saint Louis	.6583334	46.54167	120
4.	Bel Air	.6266667	48.30222	225
5.	Mont Fleuri	.5227273	50.49242	132

preserve, collapse, restore: the counts

```
. preserve

. collapse (mean) employed ilo_age (count) persons=employed ///
>     if !missing(employed), by(district)

. count
  26

. list in 1/5

     +-----------------------------------------------+
     |      district   employed    ilo_age   persons |
     |-----------------------------------------------|
  1. | English River   .6598361   42.97541       244 |
  2. |   Mont Buxton       .648      45.16       125 |
  3. |   Saint Louis   .6583334   46.54167       120 |
  4. |       Bel Air   .6266667   48.30222       225 |
  5. |   Mont Fleuri   .5227273   50.49242       132 |
     +-----------------------------------------------+

. restore

. count
  5,888
```

1. 26 rows while collapsed: one per district
2. 5,888 again after restore

COMMON ERROR

restore twice

```
. restore
nothing to restore
```

WHY

`restore` brings back the copy and then forgets it. A second `restore` has nothing to bring back.

FIX

One `preserve`, one `restore`. If you need the copy again, `preserve` again.

COMMON ERROR

Counting a text variable

```
. collapse (sum) wgt_final (count) n=unique_ind, by(region)
string variable unique_ind not allowed with statistic count
```

WHY

`unique_ind` is stored as text in this file, and `(count)` needs a number.

FIX

Count a numeric variable that is never missing for the rows you want, such as `(count) n=ilo_age`.

YOUR TURN · TASK 34

Keep the district file, get the persons back

DO

Add, under `list`, and run the block again from `preserve`:

```
save "output/district_indicators.dta", ///
  replace
restore
count
```

YOU SHOULD SEE

`file output/district_indicators.dta saved`, then 5,888: the persons are back.

IF NOT

26 instead of 5,888? `restore` did not run: run the whole block.

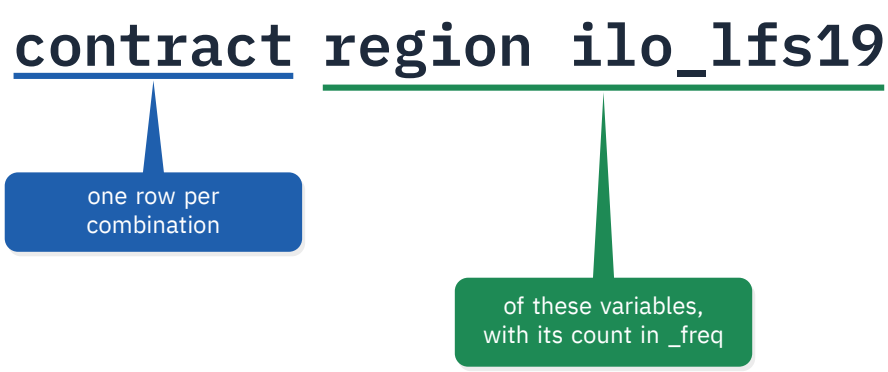
Ranking the districts (Extra of Task 33)

```
     +---------------------------------+
     |      district   employed   rank |
     |---------------------------------|
  1. |   Anse Etoile     .743295      1 |
  2. | Baie Ste Anne     .722488      2 |
  3. |  Anse Boileau    .7045454      3 |
  4. | Anse Aux Pins    .7035573      4 |
  5. |   Beau Vallon    .6964286      5 |
```

1

1. Anse Etoile: highest share employed among working-age persons interviewed, 74.3%

A frequency table, kept as data:



YOUR TURN · TASK 35

The region × status table as data

DO

Add and run the block:

```
preserve
contract region ilo_lfs19
list in 1/8

restore
```

YOU SHOULD SEE

Rows of region, status and `_freq`; the persons return at `restore`.

IF NOT

`nothing to restore`? Run the four lines together.

Your result

```
. contract region ilo_lfs19


.
. list in 1/8
```

	region	ilo_lfs19	_freq
1.	1- North	1 - Employed	508
2.	1- North	2 - Unemployed	17
3.	1- North	3 - Outside Labour Force	203
4.	1- North	.	212
5.	2- Central 1	1 - Employed	462
6.	2- Central 1	2 - Unemployed	25
7.	2- Central 1	3 - Outside Labour Force	227
8.	2- Central 1	.	167

contract with a named count, missing left out

```
. contract region ilo_lfs19, freq(n) nomiss

. list in 1/6

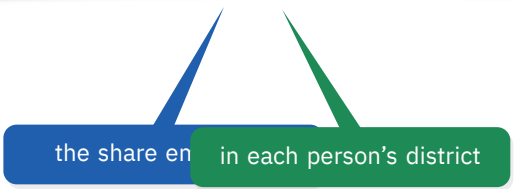
     +-----------------------------------------------+
     |       region                  ilo_lfs19     n |
     |-----------------------------------------------|
  1. |     1- North               1 - Employed   508 |
  2. |     1- North             2 - Unemployed    17 |
  3. |     1- North   3 - Outside Labour Force   203 |
  4. | 2- Central 1               1 - Employed   462 |
  5. | 2- Central 1             2 - Unemployed    25 |
     |-----------------------------------------------|
  6. | 2- Central 1   3 - Outside Labour Force   227 |
     +-----------------------------------------------+
```

1

1 `freq(n)` names the count `n`; `nomiss` drops the rows with missing status

The district value, on every person row

```
egen share_emp = mean(employed), by(district)
```



No rows are lost: `collapse` keeps one row per district, `egen` keeps every person.

YOUR TURN · TASK 36

District employment share for every person

DO

Add and run:

```
egen share_emp = mean(employed), by(district)
list district share_emp in 1/5
```

YOU SHOULD SEE

Each person carries the share of their district; neighbours share one value.

IF NOT

`variable share_emp already defined`? Run `my_day2.do` from the top.

EXTRA Do the same for the mean age: `mean(ilo_age)`.

Your result

```
. egen share_emp = mean(employed), by(district)

.
. list district share_emp in 1/5

     +-------------------------------+
     |      district   share_~p      |
     |-------------------------------|
  1. | English River   .6598361      |
  2. | English River   .6598361      |
  3. | English River   .6598361      |
  4. | English River   .6598361      |
  5. | English River   .6598361      |
     +-------------------------------+
```

The person value and the district value

```
. list district employed share_emp in 1/6

     +-------------------------------------+
     |      district   employed   share_~p |
     |-------------------------------------|
  1. | English River          0   .6598361 |
  2. | English River          0   .6598361 | 1
  3. | English River          .   .6598361 | 2
  4. | English River          1   .6598361 |
  5. | English River          0   .6598361 |
     |-------------------------------------|
  6. | English River          0   .6598361 |
     +-------------------------------------+
```

1. `employed`: this person, 0, 1 or missing
2. `share_emp`: the district, the same on every row

ON YOUR OWN · G

Where are the longest working weeks?

THE QUESTION

Mean usual hours and the number of answers, by district. Which five districts have the highest mean?

HINT

`preserve`, `collapse (mean) ... (count) ...`, rank, list the top five, `restore`.

ON YOUR OWN · G · ANSWER

```
preserve
collapse (mean) hours=ilo_job1_how_usual ///
(count) n=ilo_job1_how_usual, by(district)
egen r = rank(-hours)
sort r
list district hours n in 1/5
restore
```

```
     +-----------------------------------------------+
     |                   district      hours     n |
     |-----------------------------------------------|
  1. | La Digue and Inner islands   44.44444    63 |
  2. |              English River   42.76398   161 |
  3. |                   Bel Air   41.71631   141 |
  4. |           Ile Perseverance   41.33918   171 |
  5. |               Pointe Larue   41.18681   182 |
     +-----------------------------------------------+
```

1

1 La Digue and Inner islands: 44.4 hours, but from 63 answers only

COMMON ERROR

A typo inside by()

```
. collapse (mean) employed, by(distrct)
variable distrct not found
(error in option by())
```

WHY

The district variable is `district`; `distrct` does not exist. Stata says which option failed.

FIX

Copy variable names from the Variables window.

QUICK CHECK

You need the district share employed for a table AND every person kept for later analysis. Which tool?

A `collapse`

B `egen ..., by(district)`

C `contract`

ANSWER

You need the district share employed for a table AND every person kept for later analysis. Which tool?

A `collapse`

B `egen ..., by(district)`

C `contract`

`egen ..., by()` puts the district value on every person and keeps all rows. `collapse` would replace the persons with 26 rows.

Part 5 on one page

CHEAT CARD · KEEP THIS ONE

`preserve ... restore`	work on a copy, then get the persons back
`collapse (mean) v (count) n=v, by(g)`	one row per group
`contract a b, freq(n) nomiss`	counts of combinations, as data
`egen s = mean(v), by(g)`	the group value on every row
`egen r = rank(-v)`	rank, highest = 1

Practice on your own

PRACTICE SET · practice_D2.do, PART 5

CORE P1 One row per region: mean usual hours and the number of answers.

CORE P2 `contract ilo_sex ilo_lfs19`: how many rows, and why?

CORE P3 Mean age of the region on every person row.

EXTENSION P4 Rank the regions by share employed, highest = 1.

CHALLENGE P5 Keep only the districts with more than 200 persons aged 15 and over.

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

SAMPLE, NOT POPULATION

These shares describe the persons interviewed.

Every person counts once here. In the population, each one stands for a different number of people.

Tomorrow: the weights turn them into district estimates.

You can turn person records into district indicators, two ways.

lost? open `catchup_D2_S5.do`



6

DAY 2 · PART 6

Charts for the release

A bar for each group:

```
graph bar (mean) ilo_job1_how_usual, over(region)
```

a bar chart

the height of each bar

one bar per region

YOUR TURN · TASK 37

Mean usual hours by region

DO

Type:

```
graph bar (mean) ilo_job1_how_usual, over(region)
```

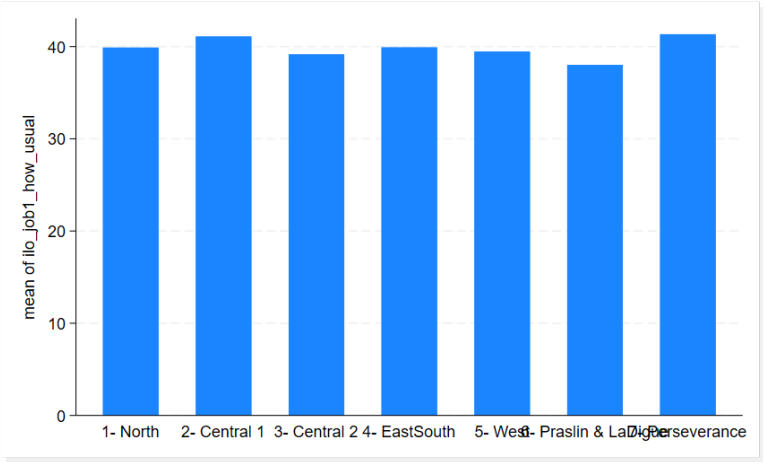
YOU SHOULD SEE

A new Graph window with seven bars, all close to 40 hours. Look at the names under the bars.

IF NOT

`variable region not found`? Open `lfs_teach.dta` again: the district file is in memory.

Your result


mean of llo_job1_how_usual
40
30
20
10
0
1- North
2- Central 1
3- Central 2
4- EastSouth
5- West
6- Praslin & La Digue
7- Perseverance

Title, axis, source

```
graph hbar (mean) ilo_job1_how_usual, over(region) ///
   title("Usual weekly hours, main job") ///
   ytitle("Mean hours") ///
   note("Source: NBS, LFS Q1 2026 (sample, unweighted)")
```

- `hbar`: bars across, so long names do not overlap
- what the chart shows
- the unit on the axis
- the source, under the chart

YOUR TURN · TASK 38

Finish the chart

DO

Add the four lines of the last slide to `my_day2.do` and run them from the Do-file Editor.

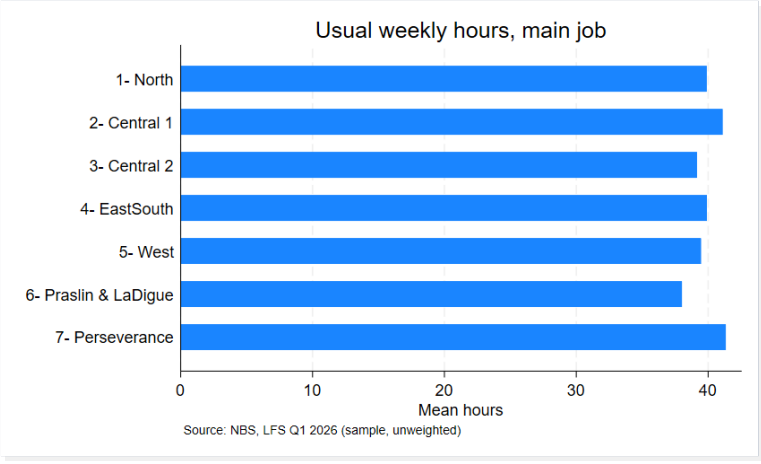
YOU SHOULD SEE

The bars now lie across, every region name readable; title on top, “Mean hours” on the axis, source below.

IF NOT

`option / not allowed`? Run the lines from the Do-file Editor, not the Command window.

Your result


Usual weekly hours, main job
1- North
2- Central 1
3- Central 2
4- EastSouth
5- West
6- Praslin & LaDigue
7- Perseverance
0
10
20
30
40
Mean hours
Source: NBS, LFS Q1 2026 (sample, unweighted)

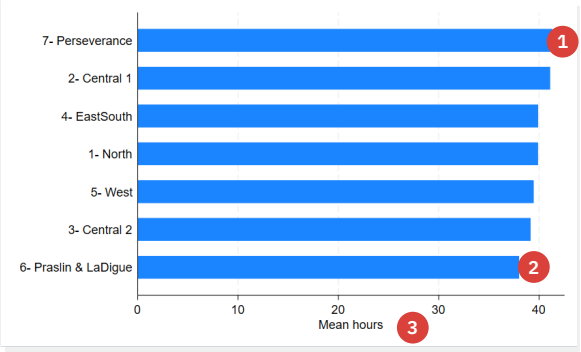
Bars in order of size

```
graph hbar (mean) ilo_job1_how_usual, over(region, sort(1) descending)
```



`sort(1)`: sort by the first statistic in the chart.

Bars in order of size, drawn



1. longest bar on top: 7- Perseverance
2. shortest at the bottom: 6- Praslin & LaDigue
3. `ytitle()` labels the axis along the bars, here the horizontal one

Save the chart as a file:

```
graph export "output/fig1_hours.png", replace
```

save the chart on screen

as this file: .png for Word, .pdf for print

overwrite last run's file

YOUR TURN · TASK 39

Export the bar chart

DO

Add directly under the Task 38 chart and run:

```
graph export "output/fig1_hours.png", replace
```

YOU SHOULD SEE

`file output/fig1_hours.png saved as PNG format`. Open it from the `output` folder.

IF NOT

`no graphs in memory`? Draw the chart first (Task 38).

EXTRA Export it again as `output/fig1_hours.pdf`.

COMMON ERROR

Exporting when no chart is open

```
. graph export "_trainer/capture/none.png", replace
could not find Graph window
```

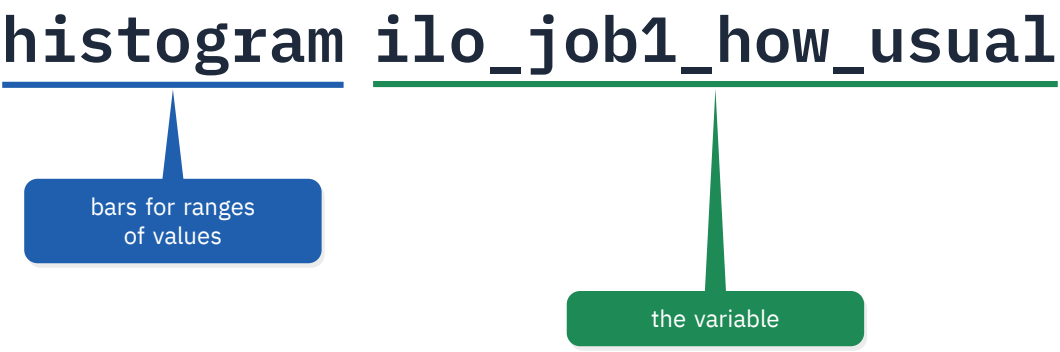
WHY

`graph export` saves the chart in the Graph window. With no chart drawn, there is nothing to save.

FIX

Put `graph export` directly under the command that draws the chart, and run them together.

The shape of one variable:



YOUR TURN · TASK 40

How are usual hours spread?

DO

Type:

```
histogram ilo_job1_how_usual
```

YOU SHOULD SEE

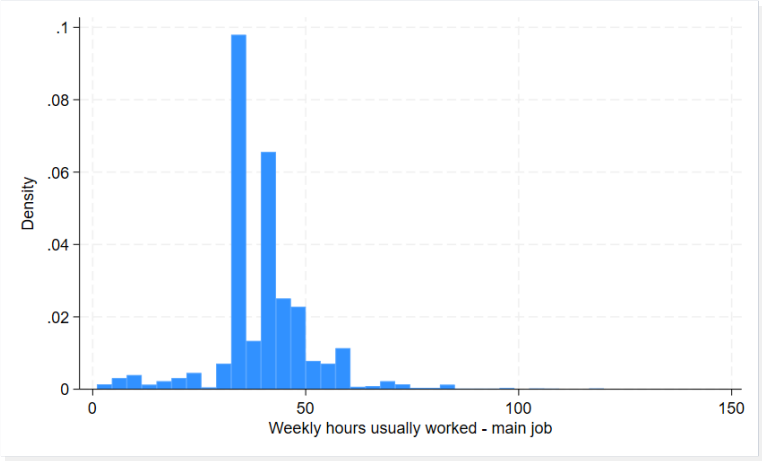
One very tall bar in the mid-30s, a second in the low 40s, and a thin tail to the right.

IF NOT

`variable not found`? Copy the name from the Variables window.

EXTRA Add `, frequency`: what changes on the y-axis?

Your result


.1
.08
.06
.04
.02
0
Density
0
50
100
150
Weekly hours usually worked - main job

QUICK CHECK

Stata says `(bin=34, start=1, width=3.5)`. What is a bin?

A. one bar: a range of 3.5 hours

B. one person

C. a missing value

ANSWER

Stata says `(bin=34, start=1, width=3.5)`. What is a bin?

A one bar: a range of 3.5 hours

B one person

C a missing value

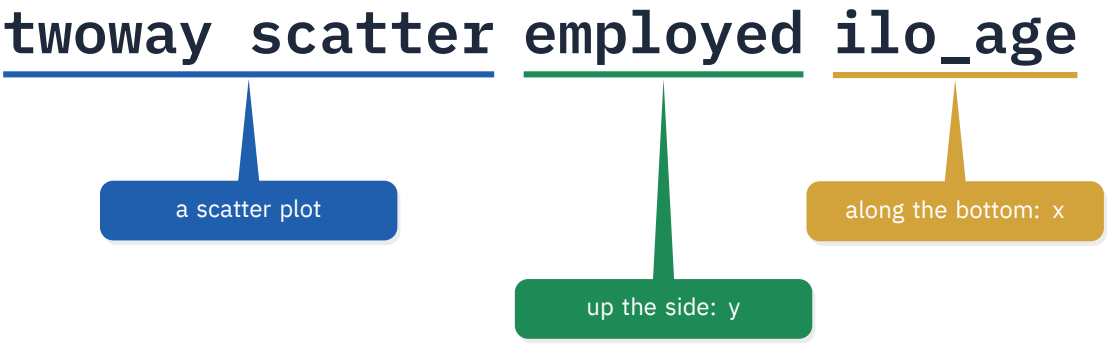
34 bars, starting at 1 hour, each 3.5 hours wide. Stata chooses them; the option `width()` changes them.

Counts, not density



1. the axis now counts persons, not density
2. the tallest bar: the 35-hour week, more than 1,000 persons
3. a thin tail up to 120 hours: the largest values of `summarize, detail`

One point for each district:



YOUR TURN · TASK 41

Share employed and mean age, by district

DO

Type:

```
use "output/district_indicators.dta", clear
twoway scatter employed ilo_age
```

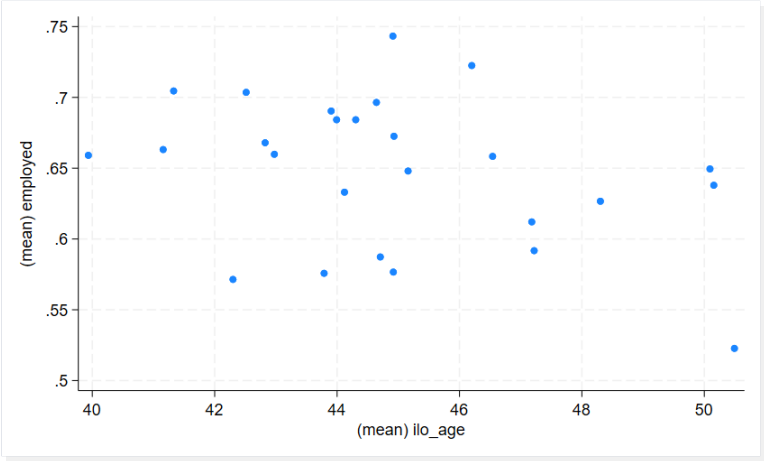
YOU SHOULD SEE

26 points: one per district.

IF NOT

`file not found`? Run Task 34 first: it saves the district file.

Your result



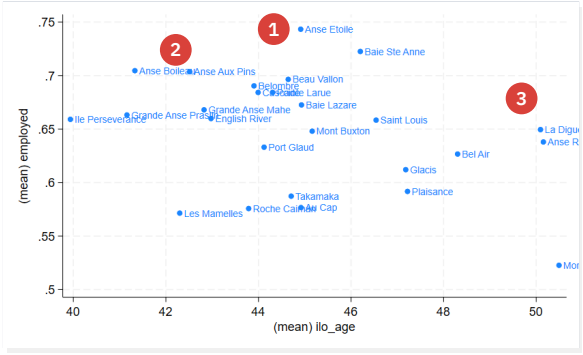
Name each point

```
twoway scatter employed ilo_age, mlabel(district)
```



With 26 districts some names overlap: label a few in a release.

Read the labelled scatter



1. Anse Etoile: the highest share employed, .743 (Task 33)
2. districts close together: their names overlap
3. a name too long for the space left is cut at the edge

ON YOUR OWN · H

Share employed, ready for the release

THE QUESTION

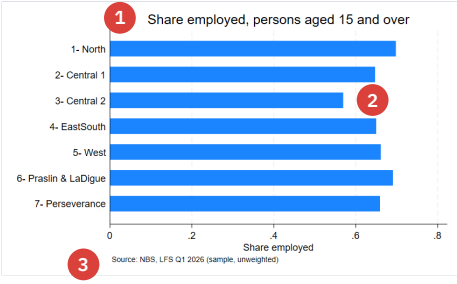
The Director wants the share employed in each region as horizontal bars, with a title, an axis title and the source under the chart.

HINT

The mean of a 0/1 variable is a share. `graph hbar (mean) employed, over(region)`, then `ytitle()`, `title()`, `note()`.

ON YOUR OWN · H · ANSWER

```
graph hbar (mean) employed, over(region) ///
ytitle("Share employed") ///
title("Share employed, persons aged 15 and over") ///
note("Source: NBS, LFS Q1 2026 (sample, unweighted)")
```



1. the title says what and for whom
2. Central 2: the shortest bar
3. the source, from `note()`

ON YOUR OWN · I

How old is the working-age population?

THE QUESTION

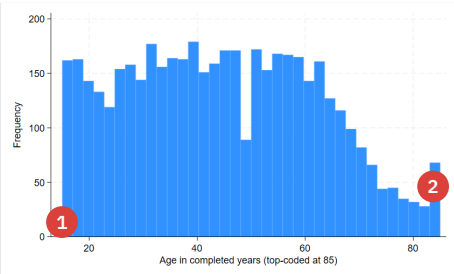
Draw the age distribution of the persons who have a labour status (15 and over), with counts on the vertical axis.

HINT

`histogram`, an `if` that keeps rows where `employed` is not missing, and `frequency`.

ON YOUR OWN · I · ANSWER

```
histogram ilo_age if !missing(employed), frequency
```



1. the first bar starts at 15: the `if` kept working-age persons only
2. the last bar jumps: everyone 85 and over is coded 85 (see the axis title)

Part 6 on one page

CHEAT CARD · KEEP THIS ONE

`graph hbar (mean) v, over(g)`	bars across, one per group
`over(g, sort(1) descending)`	longest bar first
`title() ytitle() note()`	what, unit, source
`histogram v [, frequency]`	the shape of one variable
`twoway scatter y x [, mlabel(g)]`	one point per row
`graph export "output/f.png", replace`	save the chart on screen

Practice on your own

PRACTICE SET · practice_D2.do, PART 6

CORE P1 Horizontal bars of the mean age by region, with a title and the source.

CORE P2 Histogram of usual weekly hours as counts, with bars 5 hours wide.

CORE P3 Export both charts as PNG into `output`.

EXTENSION P4 Sort the age bars from oldest to youngest region.

CHALLENGE P5 Scatter of mean usual hours against share employed, one point per district, with district names.

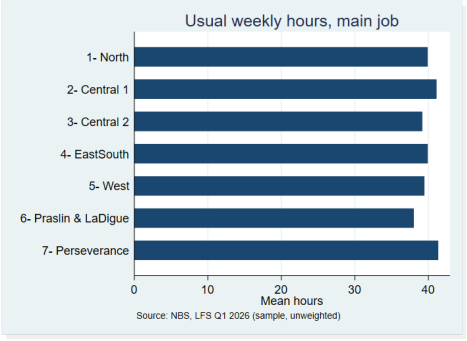
Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

STATA 18 OR NEWER

Same code, different look.

Stata 18 and newer draw with the `stcolor` scheme; Stata 17 with `s2color`. To get the older look:

```
set scheme s2color
```



Your charts carry a title, a source and a file name, ready for the release.

lost? open `catchup_D2_S6.do`



7

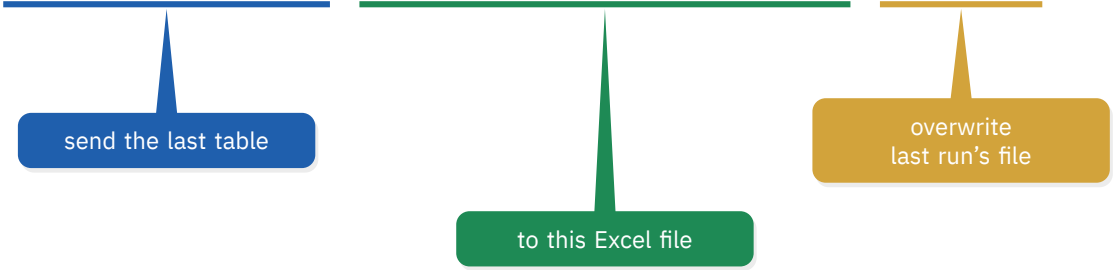
DAY 2 · PART 7

No more copy and paste

A number copied by hand is a number that can be copied wrong.

The last table, straight into Excel:

```
collect export "output/tables.xlsx", replace
```



Run it right after `table`. Stata 17 or newer.

YOUR TURN · TASK 42

Region by status, into Excel

DO

Add to `my_day2.do` and run:

```
use "data/lfs_teach.dta", clear
table (region) (ilo_lfs19)
collect export "output/tables.xlsx", replace
```

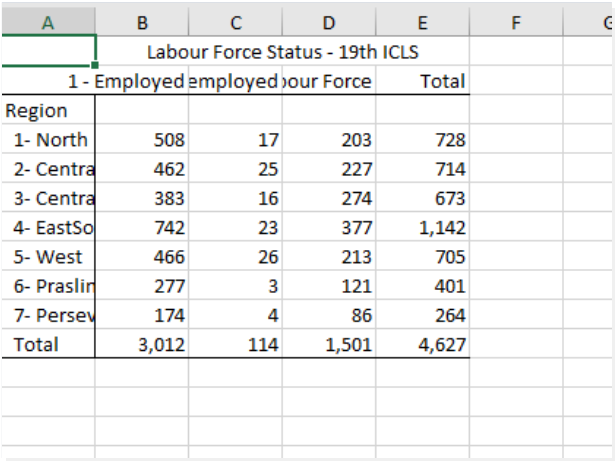
YOU SHOULD SEE

The table, then `(collection Table exported to file output/tables.xlsx)`. Open the file.

IF NOT

`file could not be saved`? The file is open in Excel: close it and run again.

Your result



	A	B	C	D	E	F	G
		Labour Force Status - 19th ICLS					
		1 - Employed	Unemployed	Labour Force	Total		
	Region						
	1- North	508	17	203	728		
	2- Centra	462	25	227	714		
	3- Centra	383	16	274	673		
	4- EastSo	742	23	377	1,142		
	5- West	466	26	213	705		
	6- Praslin	277	3	121	401		
	7- Persev	174	4	86	264		
	Total	3,012	114	1,501	4,627		

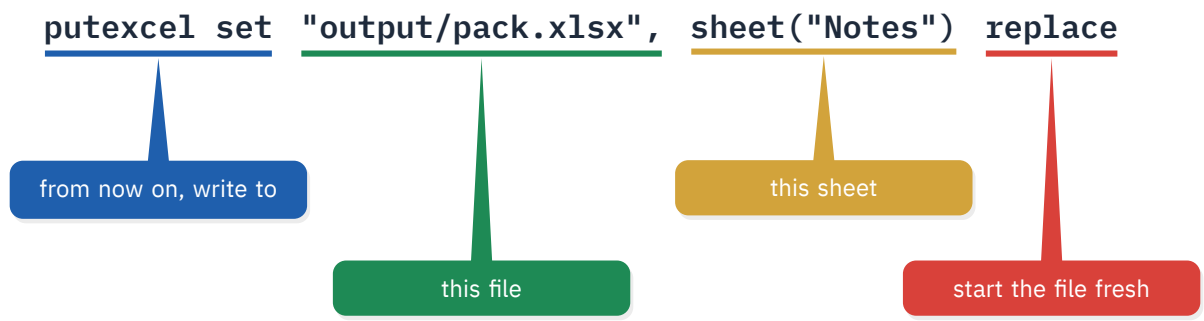
The same table, straight to Word

```
. collect export "_trainer/capture/x_tables.docx", replace
(collection Table exported to file _trainer/capture/x_tables.docx) 1
```

1. the file ending decides the format: .xlsx, .docx, .html

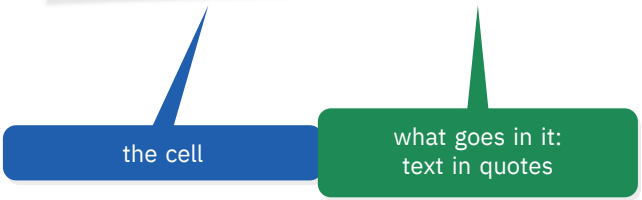
These are sample counts.
Tomorrow the same tables
become population estimates
with weights.

Choose the file and the sheet:



Write into a cell

```
putexcel A1 = "LFS Q1 2026: sample tables"
```



Several cells in one line: `putexcel A1 = "…" A2 = "…".`

YOUR TURN · TASK 43

A title and a source in the Notes sheet

DO

Add and run:

```
putexcel set "output/pack.xlsx", ///
  sheet("Notes") replace
putexcel A1 = "LFS Q1 2026: sample tables"
putexcel A2 = "Source: NBS, LFS Q1 2026 (sample)"
```

YOU SHOULD SEE

`file output/pack.xlsx saved`. Open it: the two lines in A1 and A2 of the Notes sheet.

IF NOT

`file could not be saved`? Close `pack.xlsx` in Excel and run again.

Commands leave their results behind

```
summarize ilo_job1_how_usual
return list
```

- `return list`: everything `summarize` kept
- `r(mean)`, `r(N)`, `r(min)`, `r(max)`: usable by name

The next command that leaves results replaces them.

YOUR TURN · TASK 44

A computed number in cell B3

DO

Add and run:

```
summarize ilo_job1_how_usual
return list
putexcel A3 = "Mean usual hours" B3 = (r(mean))
```

YOU SHOULD SEE

`return list` shows `r(mean)`; in the Notes sheet B3 holds that mean.

IF NOT

`r ambiguous abbreviation`? Put `r(mean)` in its own brackets: `(r(mean))`.

Read return list

```
. return list

scalars:
                  r(N) =  2999 (1)
              r(sum_w) =  2999
             r(mean) =  39.82860953651217 (2)
              r(Var) =  131.858540794452
               r(sd) =  11.48296742111777
              r(min) =  1
              r(max) =  120 (3)
              r(sum) =  119446
```

1. r(N): 2,999 answers
2. r(mean): the mean, with all its digits
3. r(sum) and the rest: all usable by name until the next command

ON YOUR OWN · J

The share employed into Excel

THE QUESTION

Write “Share employed (15+) in A4 of the Notes sheet and the share itself in B4, computed by Stata.

HINT

`summarize employed`, then one `putexcel` line with `(r(mean))`.

ON YOUR OWN · J · ANSWER

```
summarize employed
putexcel A4 = "Share employed (15+)" B4 = (r(mean))
```

```
. summarize employed

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
    employed |      4,627    .6509617    .4767176          0          1

. putexcel A4 = "Share employed (15+)" B4 = (r(mean))
file _trainer/capture/x_pack.xlsx saved
```

1. mean of a 0/1 variable = share: .651
2. written into B4

QUICK CHECK

You run `summarize`, then `tabulate`, then `putexcel B3 = (r(mean))`. What lands in B3?

A. the mean from summarize

B. nothing useful: tabulate replaced the stored results

C. the table

ANSWER

You run `summarize`, then `tabulate`, then `putexcel B3 = (r(mean))`. What lands in B3?

A the mean from summarize

B nothing useful: tabulate replaced the stored results

C the table

Every command that stores results clears the old ones. Write `putexcel` directly under `summarize`.

Your result

LFS Q1 2026: sample tables			
Source: NBS, LFS Q1 2026 (sample)			
Mean usual hours	39.82860954		

A Word document has a start and an end

```
putdocx begin
… what goes into the document …
putdocx save "output/pack.docx", replace
```

- open an empty document in memory
- write it to a file and close it

Paragraphs, text and charts

```
putdocx paragraph
putdocx text ("LFS Q1 2026: sample tables")
putdocx paragraph
putdocx image "output/fig1_hours.png"
```

- start a new paragraph
- write text into it
- place the chart exported in Task 39

The last table, into Word

```
table (region) (ilo_lfs19)
putdocx collect
```

the table you just made, as a Word table

Same idea as `collect export`, inside the document. Stata 17 or newer.

YOUR TURN · TASK 45

Build the Word pack

DO

Add and run as one block:

```
putdocx begin
putdocx paragraph
putdocx text ("LFS Q1 2026: sample tables")
putdocx paragraph
putdocx text ("Unweighted sample counts.")
table (region) (ilo_lfs19)
putdocx collect
putdocx paragraph
putdocx image "output/fig1_hours.png"
putdocx save "output/pack.docx", replace
```

YOU SHOULD SEE

`(collection Table posted to putdocx)`, then `successfully created` with the file path. Open `pack.docx`.

IF NOT

`document already open`? Type `putdocx clear` and run the block again.

COMMON ERROR

A Word document already open

```
. putdocx begin
document already open in memory
```

WHY

A run stopped after `putdocx begin` and before `putdocx save`, so the document is still open in memory.

FIX

Type `putdocx clear` once, then run the block again.

Part 7 on one page

CHEAT CARD · KEEP THIS ONE

`collect export "f.xlsx", replace`	the last table to Excel (or .docx)
`putexcel set "f.xlsx", sheet("S") replace`	choose file and sheet
`putexcel A1 = "text" B1 = (r(mean))`	text and results into cells
`return list`	what the last command stored
`putdocx begin / putdocx save "f.docx", replace`	open and save a Word document
`putdocx paragraph / text ("...") / image "f.png"`	content
`putdocx collect`	the last table into Word
`putdocx clear`	close a document left open

Practice on your own

PRACTICE SET · practice_D2.do, PART 7

CORE P1 Sex by labour force status as a table, exported to `output/sex.xlsx`.

CORE P2 The mean usual hours of women into cell B2 of a sheet called Women.

CORE P3 A Word file with a title line and the same table.

EXTENSION P4 Add the hours histogram of Part 6 to the Word file.

CHALLENGE P5 Share employed for men in B3 and for women in B4, computed by Stata.

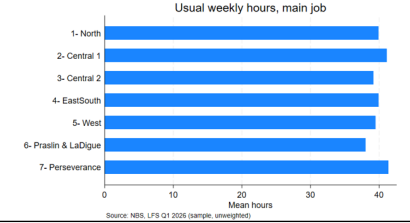
Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

Your result

LFS Q1 2026: sample tables

Unweighted sample counts.

	Labour Force Status - 19th ICLS			
	1 - Employed	2 - Unemployed	3 - Outside Labour Force	Total
Region				
1- North	508	17	203	728
2- Central 1	462	25	227	714
3- Central 2	383	16	274	673
4- EastSouth	742	23	377	1,142
5- West	466	26	213	705
6- Praslin & LaDigue	277	3	121	401
7- Perseverance	174	4	86	264
Total	3,012	114	1,501	4,627



One do-file now writes the Excel and Word pack, with no copying.

lost? open **catchup_D2_S7.do**



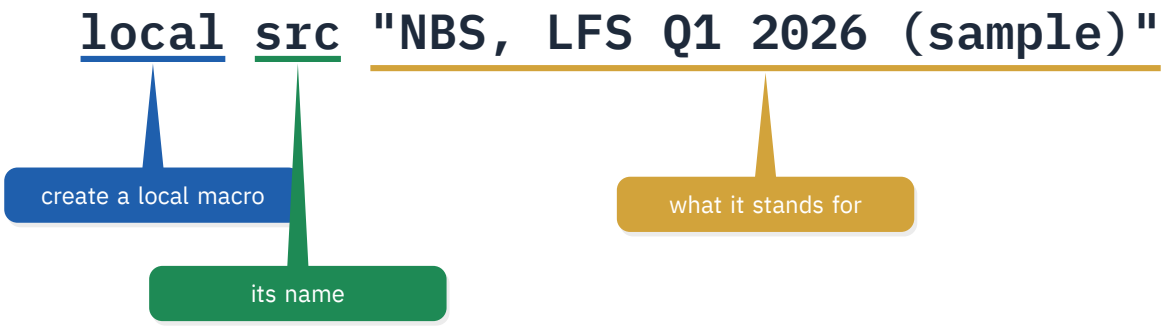
8

DAY 2 · PART 8

Write it once, run it seven times

Seven regions, seven tables: copying the code seven times invites seven chances of error.

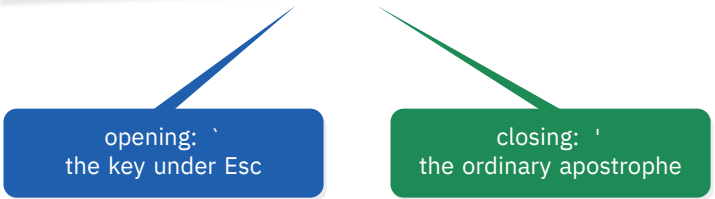
A name for a piece of text:



Use it as `` `src' ``: Stata puts the text in its place before running the line.

Two different quote marks

```
note("Source: `src'")
```



Two apostrophes, or two backticks, do not work.

YOUR TURN · TASK 46

A source line you type once

DO

Add to `my_day2.do` and run both lines together:

```
local src "NBS, LFS Q1 2026 (sample)"
histogram ilo_job1_how_usual, ///
  note("Source: `src'")
```

YOU SHOULD SEE

The histogram of Task 40 with **Source: NBS, LFS Q1 2026 (sample)** under it.

IF NOT

The note says only **Source:**? The local line did not run in the same selection.

A local, and a local with a typo

RIGHT NAME

```
. local src "NBS, LFS Q1 2026 (sample)"

. display "`src'"
NBS, LFS Q1 2026 (sample)

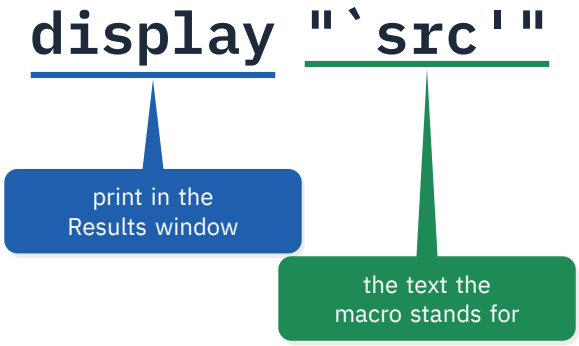
. display "Source: `src'"
Source: NBS, LFS Q1 2026 (sample)
```

TYPO: scr

```
. display "Source: `scr'"
Source:  1
```

A local that does not exist is empty, without any error: the note says only “Source:. Check the spelling when a text comes out blank.

See what a macro holds:



The same lines, for each item in a list

```
foreach v of varlist dem_mrt dem_edl mie_socpro {
    tabulate `v', missing
}
```

- v: a local that takes each name in turn
- the list: three variables, so three runs
- between the braces: what runs each time, with `v'

YOUR TURN · TASK 47

Three tables, one block

DO

Add and run all three lines together:

```
foreach v of varlist dem_mrt dem_edl mie_socpro {
  tabulate `v', missing
}
```

YOU SHOULD SEE

Three tables, one after the other, each with its missing rows.

IF NOT

`{ required`? The `{` must be the last thing on the foreach line.

EXTRA Add `agegrp` to the list.

Your result

Highest level of education	Freq.	Percent	Cum.
No schooling	50	0.85	0.85
Crèche	96	1.63	2.48
Primary year 1	98	1.66	4.14
Primary year 2	115	1.95	6.10
Primary year 3	137	2.33	8.42
Primary year 4	144	2.45	10.87
Primary year 5	122	2.07	12.94
Primary year 6	265	4.50	17.44
Secondary year 1	137	2.33	19.77
Secondary year 2	289	4.91	24.68
Secondary year 3	344	5.84	30.52
Secondary year 4	348	5.91	36.43
Secondary year 5	1,231	20.91	57.34
Advanced level	337	5.72	63.06
Vocational	947	16.08	79.14
Diploma / Higher National Diploma	412	7.00	86.14
1st Degree	177	3.01	89.15
Post graduate & higher	152	2.58	91.73
.	353	6.00	97.72
.a	125	2.12	99.85
.b	9	0.15	100.00
Total	5,888	100.00	

Employer pays social contributions	Freq.	Percent	Cum.
YES	2,243	38.09	38.09
NO	280	4.76	42.85
.	3,326	56.49	99.34
.a	39	0.66	100.00
Total	5,888	100.00	

How Stata shows a loop

```
. foreach v of varlist dem_mrt dem_edl mie_socpro {
  2.     tabulate `v', missing
  3. }

                        Marital status |      Freq.     Percent        Cum.
---------------------------------------+-----------------------------------
                                Single |      2,245       38.13       38.13
                               Married |      1,099       18.67       56.79
Living together like married partners |        986       16.75       73.54
                             Separated |         49        0.83       74.37
                              Divorced |         77        1.31       75.68
                               Widowed |        170        2.89       78.57
                                     . |      1,261       21.42       99.98
                               Refusal |          1        0.02      100.00
---------------------------------------+-----------------------------------
                                 Total |      5,888      100.00
```

1. the loop lines are echoed with numbers: 2., 3.
2. the closing brace ends the loop
3. then one table per variable, as if typed three times

Loop over the values of a variable

```
levelsof region, local(regs)
foreach r of local regs {
    … `r' …
}
```

- put the values of `region` (1 to 7) into the local `regs`
- run the block once for each value, as `` `r' ``

What levelsof finds

```
. levelsof region, local(regs)
1 2 3 4 5 6 7 (1)

. display "`regs'"
1 2 3 4 5 6 7
```

(1) the seven values of `region`, stored in the local `regs`

YOUR TURN · TASK 48

One chart per region

DO

Add and run as one block:

```
levelsof region, local(regs)
foreach r of local regs {
  graph hbar (mean) ilo_job1_how_usual ///
    if region == `r', over(district) ///
    title("Region `r'")
  graph export "output/hours_region`r'.png", ///
    replace
}
```

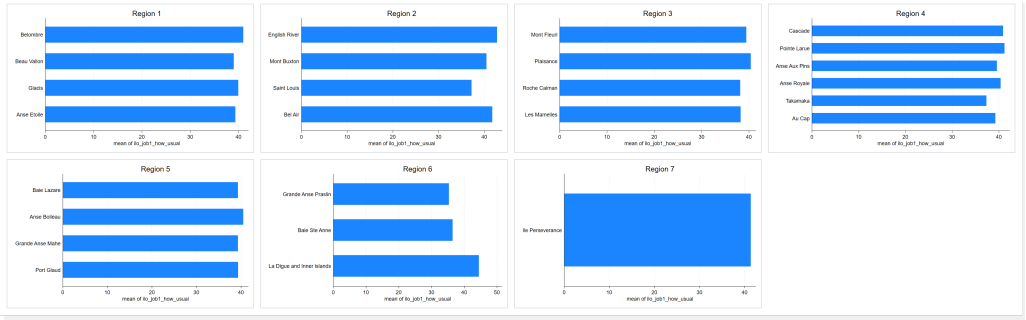
YOU SHOULD SEE

Seven charts drawn one after another, and seven `saved as PNG` lines.

IF NOT

No chart at all? Select the `levelsof` line too: `regs` exists only within one run.

Your result


Region 1
Belombre
Beau Vallon
Glacis
Anse Etoile
0
10
20
30
40
mean of ilo_job1_how_usual
Region 2
English River
Mont Buxton
Saint Louis
Bel Air
0
10
20
30
40
mean of ilo_job1_how_usual
Region 3
Mont Fleuri
Plaisance
Roche Caiman
Les Mamelles
0
10
20
30
40
mean of ilo_job1_how_usual
Region 4
Cascade
Pointe Larue
Anse Aux Pins
Anse Royale
Takamaka
Au Cap
0
10
20
30
40
mean of ilo_job1_how_usual
Region 5
Baie Lazare
Anse Boileau
Grande Anse Mahe
Port Glaud
0
10
20
30
40
mean of ilo_job1_how_usual
Region 6
Grande Anse Praslin
Baie Ste Anne
La Digue and Inner Islands
0
10
20
30
40
50
mean of ilo_job1_how_usual
Region 7
Ile Perseverance
0
10
20
30
40
mean of ilo_job1_how_usual

COMMON ERROR

The brace on the next line

```
. foreach v of varlist dem_mrt dem_edl
{ required
r(100);
```

WHY

The `{` must end the `foreach` line. On a line of its own it comes too late.

FIX

Move `{` to the end of the `foreach` line.

ON YOUR OWN · K

Two tables, one loop

THE QUESTION

With one loop, show status within age group and status within sex (row percentages). Which age group has the highest share unemployed?

HINT

`foreach v of varlist agegrp ilo_sex { ... }` with `tabulate `v' ilo_lfs19, row nofreq`.

ON YOUR OWN · K · ANSWER

```
foreach v of varlist agegrp ilo_sex {
tabulate `v' ilo_lfs19, row nofreq
}
```

```
           | Labour Force Status - 19th ICLS
 Age group | 1 - Emplo  2 - Unemp  3 - Outsi |     Total
-----------+---------------------------------+----------
     15-24 |     42.36       7.78      49.86 |    100.00
     25-34 |     84.79       2.79      12.42 |    100.00
     35-54 |     85.68       1.69      12.63 |    100.00
       55+ |     42.38       0.55      57.07 |    100.00
-----------+---------------------------------+----------
     Total |     65.10       2.46      32.44 |    100.00
```

1

1 15–24: 7.78% unemployed, three times the total

QUICK CHECK

In `foreach r of local regs`, with regs = 1 2 3 4 5 6 7, what is `` `r' `` during the third run?

A. 1

B. 3

C. 7

ANSWER

In `foreach r of local regs`, with regs = 1 2 3 4 5 6 7, what is `` `r' `` during the third run?

A 1

B 3

C 7

The local takes the items of the list in order: 1, then 2, then 3.

Part 8 on one page

CHEAT CARD · KEEP THIS ONE

`local name "text"`	a name for a piece of text
`` `name' ``	use it: backtick, name, apostrophe
`display "`name'"`	see what it holds
`foreach v of varlist a b c {`	repeat for each variable
`levelsof g, local(L)`	the values of g into a local
`foreach x of local L {`	repeat for each value
`}`	alone on the last line

Practice on your own

PRACTICE SET · practice_D2.do, PART 8

CORE P1 A local holding the source line; show it with `display`.

CORE P2 One loop: `tabulate, missing` for `ilo_sex`, `ilo_lfs19` and `region`.

CORE P3 Loop over the regions: mean age of each with `summarize`.

EXTENSION P4 Loop over the regions: one hours histogram per region, exported as PNG.

CHALLENGE P5 Loop over the regions: write each region's number of employed into column B of an Excel sheet, one row per region.

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

BRACES

{ ends the `foreach` line. } stands alone on its own line.

Anything after { on the same line, or before }, stops the loop with an error.

Indent the lines inside: the reader sees what repeats.

You wrote one block of code that produced seven charts. Day 5 builds on this.

lost? open `catchup_D2_S8.do`

Lab 2 · The tables pack

THE SITUATION

The labour statistics section wants a first sample pack: three tables in one Excel file, two charts, and a Word note. Every number from one do-file.

OPEN

`lab_D2_2.do` in `C:\NBS_Stata\day2`. Close Excel and Word before each run.

EIGHT TASKS

CORE	Tasks 1 to 6
EXTENSION	Task 7: a chart per region
CHALLENGE	Task 8: a sheet per region

CORRECT WHEN

The whole file runs with no red text; `tables_pack.xlsx` and `.docx` open.

The lab file in your editor

```
49  * ------------------------------------------------------------------
50  * TASK 2 (Core)  Table 1: persons by region and labour force status
51  *   Make the table with table, then export it to
52  *   output/tables_pack.xlsx, sheet T1. Use the options
53  *   sheet("T1") replace  so each run starts a fresh file.
54  * ------------------------------------------------------------------
55  * >>> YOUR CODE
56
57
58  * CHECK Task 2 (silent = correct)
59  import excel using "output/tables_pack.xlsx", describe
60  assert r(N_worksheet) == 1
61
62
63  * ------------------------------------------------------------------
64  * TASK 3 (Core)  Table 2: status within each region, one decimal
65  *   Same table with row percentages and nformat(%5.1f).
66  *   Export it to the same file, sheet T2. This time use
67  *   sheet("T2") modify  so sheet T1 is kept.
68  * ------------------------------------------------------------------
69  * >>> YOUR CODE
70
71
72  * CHECK Task 3 (silent = correct)
73  import excel using "output/tables_pack.xlsx", describe
74  assert r(N_worksheet) == 2
```

Green note when your last check is silent. Then the Challenge task.

Day 2, done

YOU CAN NOW

- join files by rows and by key
- turn tables between long and wide
- build release tables with `table`
- make district indicators
- draw and export charts
- write Excel and Word from a do-file
- repeat code with a loop

TODAY'S PRODUCT

```
pack.xlsx
pack.docx
```

TOMORROW

Weights, PSUs and strata: the same tables as official estimates, with their margins of error.

One product a day

Day 5 · one-button release

Day 4 · analytical note

Day 3 · official estimates

Day 2 · tables pack — done

Day 1 · clean analysis file — done

Daily check

1. Open `check_D2.do` from the `day2` folder.
2. Type your **participant code** on the first line.
3. One sentence: what was unclear today?
4. Close Excel and Word. Save and run with **Ctrl + D**.

```
.
. display as text _n "`done' of 6 checks done. Thank you."

6 of 6 checks done. Thank you.

. display as text "Your result file: output/check_D2_`code'.csv"
Your result file: output/check_D2_P00.csv

.
end of do-file
```

Read the last line: how many of 6 checks are done. It is not a grade.

Tomorrow the tables you built today become NBS's official figures.